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PRIMARY RESEARCH PAPER

Seasonal dynamics of copepod functional traits as ecological indicators in the marine ecosystems

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Abstract We investigated the ecological indicators and functional traits of copepods across the East Sea (ES), South Sea (SS), Yellow Sea (YS), and East China Sea (ECS) over four seasons in 2020. A total of 128 copepod species were identified and categorized into four functional traits based on variations in body length, feeding type, trophic group, and spawning strategy. In the YS, smaller copepods, primarily omnivores and mixed feeders, dominated spring communities, influenced by cooler temperatures and the Yellow Sea Cold Water Mass. During summer, the SS + ECS regions, affected by the nutrient-rich Kuroshio Current, saw an increase in larger omnivorous-herbivores. Autumn featured a diverse mix of sac-spawners and omnivorous-carnivores across all regions, while winter communities showed varied feeding strategies, with the ES exhibiting stable species composition and the YS and SS + ECS adapting to colder, nutrient-limited conditions. The SS + ECS consistently displayed higher species richness and functional diversity, driven by complex oceanographic processes and monsoonal patterns. Distance-based Redundancy Analysis (dbRDA) revealed significant correlations between multiple environmental variables, with the spatial distribution of indicator species. Threshold Indicator Taxa analysis (TITAN) further indicated species-specific threshold responses to hydrographic changes, emphasizing the role of temperature and salinity in structuring copepod communities across different regions. This study underscores the importance of regional

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and seasonal variability in marine ecosystems and the utility of copepod functional traits as indicators of environmental change.

Keywords Copepoda · Marine ecosystems · Functional traits · Korean peninsula · Biodiversity

Introduction

Zooplankton play a crucial role in marine food webs, facilitating biogeochemical cycles and energy transfer from primary producers to higher trophic levels (Banse, 1995; Beaugrand et al., 2003; Ducklow et al., 2001; Giering et al., 2014; Jónasdóttir et al., 2015). They are essential as a food source for fish and influence the growth and recruitment of juvenile fish (Meeren et al., 2008). Particularly, the composition and abundance of copepod species that dominate zooplankton communities directly affect the direction and efficiency of these cycles and energy flows in aquatic ecosystems (Mitra et al., 2014; Steinberg & Landry, 2017; Brun et al., 2019). Moreover, the sensitivity of copepods to environmental factors such as temperature and salinity makes them reliable indicators of climate change (Taylor et al., 2002; Mackas et al., 2007; Richardson, 2008). Studies of diverse marine environments worldwide, including the South Atlantic, Mediterranean, and Arctic Oceans, have highlighted the role of copepods as indicators of ecological and climatic stress, as well as the importance of understanding copepod distribution and functional diversity (Bradford-Grieve et al., 2010; Benedetti et al., 2018; McGinty et al., 2018, 2021; Vilgrain et al., 2021).

One of the fundamental challenges in marine ecology is identifying robust bioindicators that can effectively signal environmental changes and ecosystem health (Borja, 2014; Bonnet & Frid, 2004). Copepods, due to their short life cycles, rapid responses to environmental variability, and functional diversity, serve as key bioindicators in marine ecosystems (Steinberg & Landry, 2017; Tang et al., 2022; Choo et al., 2023). However, most traditional bioassessment studies have relied on species composition and abundance metrics, which may not fully capture ecosystem shifts (Barton et al., 2013; Li et al., 2022). Changes in copepod community composition have been linked to large-scale oceanographic shifts, such as warming-induced poleward migrations of *Calanus finmarchicus* (Gunnerus, 1770) and *Calanus helgolandicus* (Claus, 1863) in the North Atlantic (Strand et al., 2020; Falkenhaug et al., 2022). Local temperature changes and latitudinal temperature variability provide important selective pressures for zooplankton adaptation. While annual temperature variability in polar and tropical regions is only 1–3 °C, it reaches 5–10 °C in temperate regions (Clarke & Gaston, 2006). Such variability strongly influences the type of thermal adaptation observed within a given region, determining whether species exhibit stenothermal or eurythermal characteristics (Sunday et al., 2011). Climate change affects not only the ecological responses of zooplankton but also their phenotypic traits. Recent studies indicate that shifts in zooplankton emergence timing reflect both physiological plasticity and potential evolutionary selection. For instance, the advancement of seasonal emergence has been interpreted as a response to warming temperatures (Richardson, 2008). Additionally, reductions in zooplankton body size (Pitois & Fox, 2006; Daufresne et al., 2009) suggest an adaptive trade-off favoring rapid development over large body size, a pattern commonly observed under warming conditions. Functional traits such as body size, reproductive strategy, and trophic interactions not only mediate species survival and competition but also serve as indicators of ecosystem shifts (Benedetti et al., 2018).

The seas surrounding the Korean Peninsula, including the East Sea (ES) (also known as the Japan Sea), South Sea (SS), Yellow Sea (YS), and East China Sea (ECS), are each influenced substantially by the complex interplay of regional and global oceanic currents, thus displaying unique physical and biological characteristics (Park et al., 2013; Yoon et al., 2016; Liu et al., 2021; Wang et al., 2003). In the ECS, the interplay of major currents such as the Taiwan Warm Current (TW), Tsushima Warm Current (TWC), and Kuroshio Current (KC), along with the influence of the Changjiang diluted water (CDW), creates a complex environment that influences copepod distribution and ecosystem processes (Chen & Wang, 1999; Lie & Cho, 2016; Liu et al., 2021). The hydrological conditions in the YS, especially the Yellow Sea Cold Water Mass (YSCWM) and the Yellow Sea Warm Current (YSWC), exhibit distinct seasonal changes and significantly influence marine ecosystem dynamics (Zhang et al., 2008). The YSCWM, which

forms in spring, strengthens in summer, and declines in autumn, plays a crucial role in providing a refuge for temperate copepod species from the high temperatures of summer (Zhang et al., 2008; Wang et al., 2003). The YSWC, mainly occurring in winter, is the sole current transporting high-salinity and high-temperature water from the TSWC to the YS (Xu et al., 2009). The surface current circulation and physical oceanographic thermohaline circulation system of the ES resemble those of large oceans, and therefore, the ES is often referred to as a "Miniature Ocean" (Park et al., 2013; Yoon et al., 2016). Surrounded by the Korean Peninsula, Russia, and the Japanese archipelago, this semi-enclosed sea of the northwest Pacific Ocean is known for its high productivity, particularly in summer, due to the shallow surface mixed layer that facilitates the upward supply of nutrients (Kwak et al., 2013). The SS is notably affected by the TSWC, a branch of the KC, as well as the diluted waters from the Yangtze River, also known as the Changjiang diluted water (Lie & Cho, 2016). During warming periods, the SS is particularly influenced by the effects of cold water masses from both the YS and the ES due to winter mixing, including both the YSCWM and the cold water currents of the ES (Zhong et al., 2018). Furthermore, the SS is significantly influenced by coastal waters from Korea (Jung et al., 2012).

Beyond the described seasonal dynamics, broader climatic phenomena like the El Niño-Southern Oscillation (ENSO) and Pacific Decadal Oscillation (PDO) drive variability over interannual to decadal timescales (Beaufort & Grelaud, 2017). ENSO events disrupt upwelling processes, leading to shifts in nutrient availability and subsequent changes in primary productivity, which cascade through trophic levels, affecting copepod populations (Chavez et al., 2003; Hooff & Peterson, 2006; Beaufort & Grelaud, 2017). On a longer scale, PDO phases influence extended patterns of sea surface temperatures, altering zooplankton community dynamics, including abundance and phenology (Mackas et al., 2012, 2013). The intersection of these natural cycles with anthropogenic climate change has amplified their effects, increasing unpredictability in marine productivity and altering biogeochemical cycles significantly (Beaugrand et al., 2010; Benedetti et al., 2018; York, 2018). Human-induced pollution is further altering these natural cycles, particularly through coastal eutrophication and chemical contamination. Increased nutrient input from terrestrial sources has led to fluctuations in primary production, while chemical pollutants impact copepod community structure (Malzahn et al., 2010; Harfmann et al., 2019). These changes suggest that contemporary ecosystems are not only subject to natural variability but are also undergoing fundamental transformations due to increasing anthropogenic stressors.

The marine environment surrounding Korea exhibits unique zooplankton distribution patterns driven by oceanographic factors such as temperature, salinity, and climatic phenomena like the Asian summer monsoon and heatwaves (Kim et al., 2024). However, previous studies focusing on local zooplankton distributions have been limited by spatial and temporal constraints, often focusing on specific seasons or small regions, which restricts the broader understanding of zooplankton variability (Shi et al., 2020; Kim & Kang, 2020). Most previous research has focused primarily on winter and summer (Lee et al., 2019; Choo et al., 2023), with limited attention given to transitional seasons such as spring and autumn, which are critical periods for ecosystem shifts. Furthermore, the spatial constraints of past studies have restricted comprehensive assessments of regional hydrographic influences on copepod communities. Addressing these limitations by expanding research to encompass a wider geographical range and multiple seasons is crucial for gaining a more nuanced understanding of copepod community dynamics and their responses to oceanographic variability. Such research would provide more comprehensive insights into zooplankton distribution patterns in Korean waters and their broader ecological implications.

Zooplankton research has traditionally relied on morphological taxonomy. However, a growing number of studies have recently recognized that this method provides limited insights into the functional roles and responses of species within ecosystems (Barton et al., 2013; Mouillot et al., 2013; Li et al., 2022). Moreover, recent studies have shifted their focus toward the functional traits of copepods, including their morphological, biochemical, physiological, structural, phenological, and behavioral characteristics (Violle et al., 2007; Pomerleau et al., 2015; Sodré & Bozelli, 2019), which play a significant role in the fitness of organisms and ecosystem dynamics (Record et al., 2013; McGinty et al., 2018). The functional approach categorizes

species into groups based on ecological roles, offering clearer insights into their environmental responses compared to traditional taxonomy based on morphological features (Violle et al., 2007; Gagic et al., 2015; Li et al., 2022; Tang et al., 2022). The study of functional diversity, a crucial component of biodiversity that encompasses a variety of functional traits within a biological assembly, offers crucial insights into how these traits relate to ecosystem functions (Petchey & Gaston, 2002; Tilman, 2001). Furthermore, the structure, functions, and services of marine ecosystems are steadily undergoing significant changes due to climate change and human activities, affecting both high-latitude and temperate marine areas (Beaugrand et al., 2010; Schmidt et al., 2020; McGinty et al., 2021). Therefore, diverse approaches are needed to understand the relationship between biodiversity and ecosystem functioning, with studies on functional traits providing greater explanatory power for ecosystem functioning than species richness or abundance alone (Violle et al., 2007; Gagic et al., 2015; Li et al., 2022).

To address existing gaps in the spatial and seasonal characterization of zooplankton communities, this study investigates the functional traits of copepods as ecological indicators across four marginal seas surrounding the Korean Peninsula. We hypothesize that (1) copepod community structure and functional trait composition vary markedly among the ES, SS + ECS, and YS, reflecting distinct hydrographic regimes; (2) seasonal fluctuations in environmental drivers particularly water temperature, salinity, and chlorophyll-*a* drive consistent shifts in dominant traits, including body size, reproductive strategy, and feeding mode; and (3) taxa identified through functional trait analysis and threshold indicator taxa analysis display region specific ecological thresholds, indicative of local adaptation to hydrographic gradients. By integrating trait-based bioindicator assessments with quantitative threshold detection, this study seeks to advance predictive frameworks for interpreting ecosystem responses to accelerating climatic and oceanographic change.

Materials and methods

Study area and zooplankton sampling

To conduct a comprehensive study on the marine environment and zooplankton distribution characteristics around the Korean Peninsula, a seasonal survey was conducted at 60 stations in 2020 using the National Institute of Fisheries Research's marine research vessel (Figs. 1, 2). The study seasons were delimited as follows: February 2020 (Winter), April–May 2020 (Spring), August 2020 (Summer), and October–November 2020 (Autumn). The vertical distribution of water temperature (°C) and salinity was measured with a calibrated SBE 9/11 CTD (Sea-Bird Electronics, Bellevue, WA, USA), focusing on the data collected during the descent phase. For the analysis of chlorophyll (Chl)-*a* concentration (μg l^{-1}), water samples were collected at multiple standard depths using a Niskin bottle (8 L PVC) attached to the CTD/rosette sampler. Although our study focused mainly on seven standard depths (0, 10, 20, 30, 50, 75, and 100 m), the sampling extended to deeper layers, including 125, 200, and up to 500 m in certain stations, depending on the depth of each specific location. The filtered samples were preserved at − 80 °C in 15-ml tubes wrapped in aluminum foil. After the filters were placed in 90% aqueous acetone for 24 h, the Chl-*a* concentration of the extract was measured using a fluorometer (Turner Designs 10-AU Fluorometer, Sunnyvale, CA, USA) calibrated with a Chl-*a* standard (Sigma-Aldrich, Darmstadt, Germany).

Zooplankton samples were collected by vertical tows 5 m from the bottom to the surface layer at the station at a towing speed of approximately 1 m s^{-1} using a conical zooplankton net (mouth opening 60 cm, mesh size 200 μm) equipped with a flowmeter (model 438,115; Hydro-Bios co., Altenholz, Germany). After collection, the zooplankton samples were split into various ratios ranging from 1:2 to 1:512 using a Folsom-type plankton separator until the final population reached approximately 300 individuals. Species identification was conducted under an optical microscope (Model SMZ645; Nikon, Tokyo, Japan), with a focus on adult copepods. This strategic focus was chosen to provide deeper insights into the adult segment of the copepod community, considering their pivotal roles in marine ecosystems. For instances requiring enhanced detail for accurate

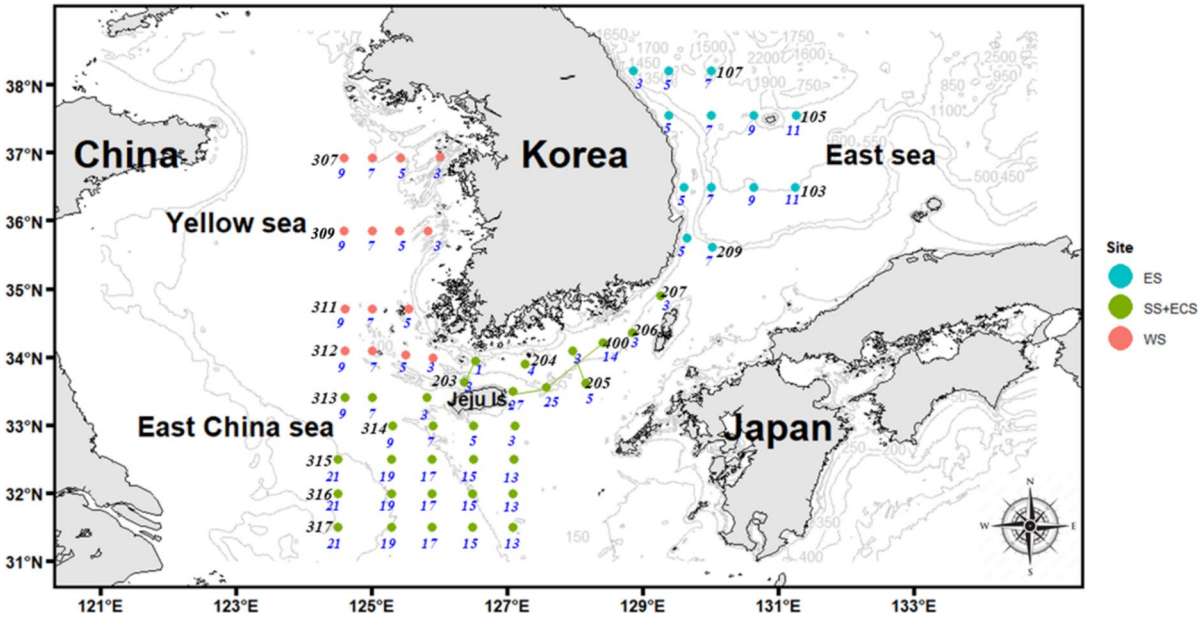

Fig. 1 Map of the sampling stations in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS). Sampling stations are indicated by colored symbols: blue circles for the East Sea (ES), green circles for the South Sea + East China Sea (SS + ECS), and red circles for the Yellow Sea (YS). Station numbers are labeled near each location for reference. The numbers represent individual sampling site

species differentiation, specific characteristic appendages of the zooplankton were dissected and observed on a glass slide under a high-power optical microscope (Model ECLIPSE 80i; Nikon, Tokyo, Japan), with a magnification range of 40 × to 400 ×. The zooplankton counts were then standardized to express abundance as the number of individuals per cubic meter (ind. m^{-3}) of water. The copepod species were identified based on the previous studies by Chihara & Murano (1997), Soh et al. (2013), and Soh and Moon (2014).

Indicator species

In this study, the indicator value index (IndVal) of species was used to express the importance of a given species as an ecological indicator within the copepod community (Dufrene & Legendre, 1997). The IndVal index is a measure that combines two elements: group specificity and group fidelity. Group specificity (A_ij) is the ratio of the average abundance of a species in a particular group to its total average abundance across all groups. Group fidelity (B_ij), on the other hand, is the proportion of sites within a group where the species is present. The overall IndVal index for a species 'i' in a group 'j' is then calculated by multiplying these two components and scaling the result to a percentage: IndVal_ij = (A_ij × B_ij) × 100. To calculate the indicator value, the 'titan' function in the 'TITAN2' package was used (Baker et al., 2023). The permutation number was set to 500, and bootstrap resampling was set to 500. The extracted results maintain reliability of 0.95 or higher. Species individual IndVal of 29% or higher and IndVal p value less than 0.05 were considered for final selection.

Functional traits of indicator species

The functional traits of indicator species were extracted based on Indicator species analysis and cluster assemblages by each season and were divided according to Benedetti et al. (2016) as follows:

i. Life history traits related to reproduction and spawning strategies: broadcasters, sac-spawners.
ii. Morphological traits were based on body length ranges and their ecological relevance (Nandini

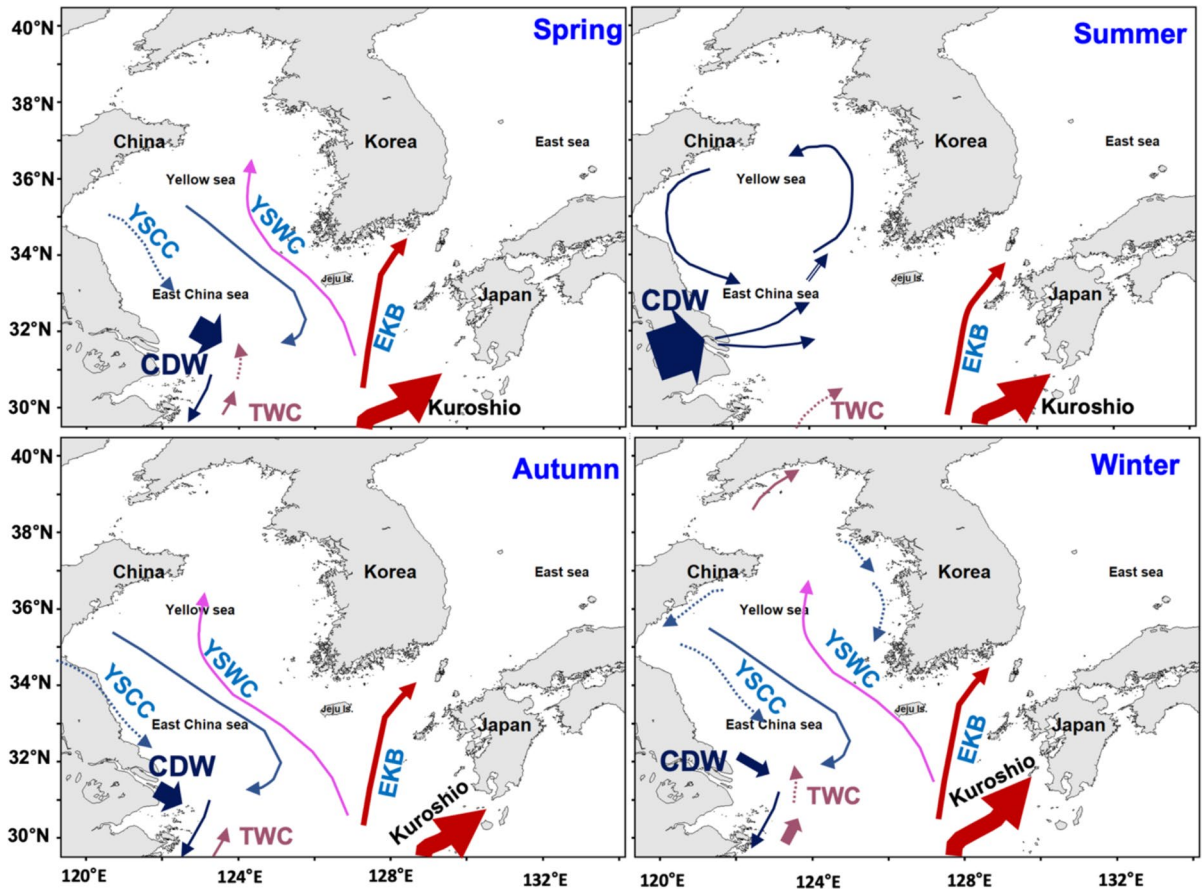

Fig. 2 Seasonal ocean current patterns in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS). The arrows indicate schematic representations of observed ocean currents around the Korean Peninsula based on the previous studies (Xiong et al., 2020). Abbreviations: Yellow Sea Coastal Current (YSCC), Yellow Sea Warm Current (YSWC), Changjiang Diluted Water (CDW), Eastern Kuroshio Branch (EKB), Tsushima Warm Current (TWC)

& Sarma, 2003; Vogt et al., 2013): body length (mm), < 1, 1–2, 2–3, > 3 mm.

iii. Physiological traits related to trophic groups (Pomerleau et al., 2015; Benedetti et al., 2016): carnivores, omnivores–herbivores (O–H), omnivores–carnivores (O–C), omnivores–detritivores (O–D), and NA. Each class was defined as follows: carnivores are exclusively predatory; O–H feed mainly on phytoplankton and occasionally on other organisms; O–C feed on organisms and sometimes on phytoplankton and organic matter; O–D feed on aggregated organisms such as marine snow, including carcasses and particulate organic matter; NA represents unidentified trophic traits.

iv. Behavioral traits related to feeding type (Kiørboe, 2011): ambush feeding, cruise feeding, current feeding, and mixed feeding (species definitions can vary depending on changes in feeding strategies).

Data analysis

To identify the characteristics of copepod communities according to season, hierarchical clustering analysis was performed according to Ward's link method based on Bray–Curtis dissimilarity. Non-metric multi-dimensional scaling (NMDS) analysis was conducted to visualize the similarities among

observation stations based on copepod abundance, following the methodology described by Oksanen et al. (2022), and the results are presented in Supplementary Fig. 1. To determine whether there were significant differences in environmental variables among locations, a one-factor Permutational Analysis of Variance (PERMANOVA) and pairwise comparison test with Bonferroni correction were performed, using significance levels of 0.05 and 0.001.

Next, ecological indices were calculated to examine the copepod community structure. Particularly, Margalef's richness index (D) was calculated to assess species richness, whereas Pielou's evenness index (J′) was used to assess the distribution of individuals among different species. Simpson's dominance index (λ) was used to evaluate the probability of two randomly selected individuals belonging to the same species. The Shannon–Weaver diversity index (H′) was applied for diversity and evenness estimation. Each of these indices was log-transformed (log $(x+1)$) to stabilize variances and improve normality, facilitating the application of parametric statistical analyses where applicable. However, for variables that did not meet normality assumptions even after transformation, nonparametric Kruskal–Wallis H tests were conducted to assess the significance of observed changes in ecological indices and environmental variables across different seasons and coastal waters of Korea (including the ES, SS + ECS, and YS).

The distance-based redundancy analysis (dbRDA) was conducted to explore the relationships between environmental variables and species distribution patterns. This analysis specifically focused on the IndVal species identified as key ecological indicators in previous studies. The dbRDA was performed separately for each season to identify station groups that included indicator species responding to varying environmental conditions, such as temperature, salinity, and Chl-*a* concentration. Data processing, statistical analyses, and visualization were conducted using the 'tidyverse' (Wickham et al., 2019), 'reshape2' (Wickham, 2007), 'vegan' (Oksanen et al., 2022), 'pairwiseAdonis' (Martinez, 2017), 'factoextra' (Irnawati et al., 2020), and 'ggplot2' (Wickham, 2016) packages within the R programming environment (R Core Team, 2024).

Results

Environmental variables

The depth-averaged environmental variables across the ES, SS + ECS, and YS exhibited significant temporal and spatial variations, particularly in water temperature, salinity, and Chl-*a* concentrations (Table 1; Fig. 3). In spring, water temperatures were significantly higher in the SS + ECS (15.61 ± 1.51 °C, $p < 0.05$) compared to the ES (10.40 ± 1.98 °C) and YS (9.02 ± 1.22 °C), with the YS exhibiting the lowest salinity (32.51 ± 0.35, $p < 0.05$) and the highest Chl-*a* concentrations (2.17 ± 1.57 μg l^{-1}, $p < 0.05$). The trend continued into summer, with the SS + ECS presenting the highest water temperatures (21.60 ± 1.71 °C, $p < 0.05$) and the YS the lowest salinity (32.10 ± 0.53, $p < 0.05$), along with notably higher Chl-*a* concentrations (0.94 ± 0.84 μg l^{-1}, $p < 0.05$). Similar patterns were observed in autumn, with the SS + ECS and YS showing higher Chl-*a* concentrations (0.73 ± 0.25 and 0.87 ± 0.64 μg l^{-1}, respectively, $p < 0.05$). In winter, despite an overall water temperature drop, the SS + ECS still exhibited relatively higher temperatures (14.55 ± 2.20 °C, $p < 0.05$) and Chl-*a* concentrations (1.36 ± 1.44 μg l^{-1}). Although the differences in Chl-*a* concentrations were not statistically significant among the three regions ($p > 0.05$), the SS + ECS showed relatively higher values than both the ES (0.68 ± 0.39 μg l^{-1}) and YS (0.58 ± 0.27 μg l^{-1}).

Spatiotemporal distribution of copepods

The seasonal variations in copepod abundance exhibited distinct patterns across the ES, SS + ECS, and YS (Fig. 4). The average abundance was highest in YS during summer (1011 ± 626 ind. m^{-3}, $p < 0.01$) and lowest in ES during spring (212 ± 118 ind. m^{-3}, $p < 0.05$). In spring, copepod abundance followed the order of YS > SS + ECS > ES, but YS exhibited significantly higher abundance than ES and SS + ECS ($p < 0.05$), but ES and SS + ECS did not differ significantly ($p > 0.05$). In summer, YS exhibited a significantly higher abundance than both ES and SS + ECS ($p < 0.01$), while ES and SS + ECS did not differ significantly ($p > 0.05$). In autumn, SS + ECS recorded the highest abundance (451 ± 272 ind. m^{-3}), but no significant differences were observed among the

Table 1 Seasonal variation of depth-averaged water temperature (°C), salinity, and Chlorophyll-*a* concentrations (Chl-*a*, μg l^{-1}) in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS)

Season	Environmental variables	ES	SS + ECS	YS
Spring	Water temperature (°C)	10.40 ± 1.98^{a} ($\chi^2 = 43.71$, df = 2, **)	15.61 ± 1.51^{b} ($\chi^2 = 43.71$, df = 2, **)	9.02 ± 1.22^{a} ($\chi^2 = 43.71$, df = 2, **)
	Salinity	34.29 ± 0.10^{b} ($\chi^2 = 26.19$, df = 2, **)	33.62 ± 0.87^{a} ($\chi^2 = 26.19$, df = 2, **)	32.51 ± 0.35^{a} ($\chi^2 = 26.19$, df = 2, **)
	Chl-*a* (μg l^{-1})	0.98 ± 0.39^{a} ($\chi^2 = 6.96$, df = 2, *)	1.15 ± 0.69^{a} ($\chi^2 = 6.96$, df = 2, *)	2.17 ± 1.57^{b} ($\chi^2 = 6.96$, df = 2, *)
Summer	Water temperature (°C)	13.48 ± 2.79^{a} ($\chi^2 = 40.08$, df = 2, **)	21.60 ± 1.71^{b} ($\chi^2 = 40.08$, df = 2, **)	18.54 ± 1.97^{a} ($\chi^2 = 40.08$, df = 2, **)
	Salinity	33.74 ± 0.21^{b} ($\chi^2 = 29.81$, df = 2, **)	32.26 ± 1.29^{a} ($\chi^2 = 29.81$, df = 2, **)	32.10 ± 0.53^{a} ($\chi^2 = 29.81$, df = 2, **)
	Chl-*a* (μg l^{-1})	0.35 ± 0.17^{a} ($\chi^2 = 12.41$, df = 2, **)	0.58 ± 0.31^{a} ($\chi^2 = 12.41$, df = 2, **)	0.94 ± 0.84^{b} ($\chi^2 = 12.41$, df = 2, **)
Autumn	Water temperature (°C)	13.12 ± 2.44^{a} ($\chi^2 = 43.22$, df = 2, **)	20.38 ± 1.05^{b} ($\chi^2 = 43.22$, df = 2, **)	18.17 ± 0.94^{a} ($\chi^2 = 43.22$, df = 2, **)
	Salinity	33.57 ± 0.12^{b} ($\chi^2 = 26.04$, df = 2, **)	33.47 ± 0.78^{a} ($\chi^2 = 26.04$, df = 2, **)	32.08 ± 0.47^{a} ($\chi^2 = 26.04$, df = 2, **)
	Chl-*a* (μg l^{-1})	0.34 ± 0.13^{a} ($\chi^2 = 23.40$, df = 2, **)	0.73 ± 0.25^{a} ($\chi^2 = 23.40$, df = 2, **)	0.87 ± 0.64^{b} ($\chi^2 = 23.40$, df = 2, **)
Winter	Water temperature (°C)	10.44 ± 1.10^{a} ($\chi^2 = 36.49$, df = 2, **)	14.55 ± 2.20^{b} ($\chi^2 = 36.49$, df = 2, **)	9.12 ± 1.91^{a} ($\chi^2 = 36.49$, df = 2, **)
	Salinity	34.21 ± 0.07^{b} ($\chi^2 = 24.63$, df = 2, **)	33.96 ± 0.80^{a} ($\chi^2 = 24.63$, df = 2, **)	32.65 ± 0.53^{a} ($\chi^2 = 24.63$, df = 2, **)
	Chl-*a* (μg l^{-1})	0.68 ± 0.39^{a} ($\chi^2 = 4.40$, df = 2, ns)	1.36 ± 1.44^{a} ($\chi^2 = 4.40$, df = 2, ns)	0.58 ± 0.27^{a} ($\chi^2 = 4.40$, df = 2, ns)

Values are presented as mean ± standard deviation. Different superscript letters (a, b) indicate statistically significant differences among regions within the same season (*$p < 0.05$, **$p < 0.01$)

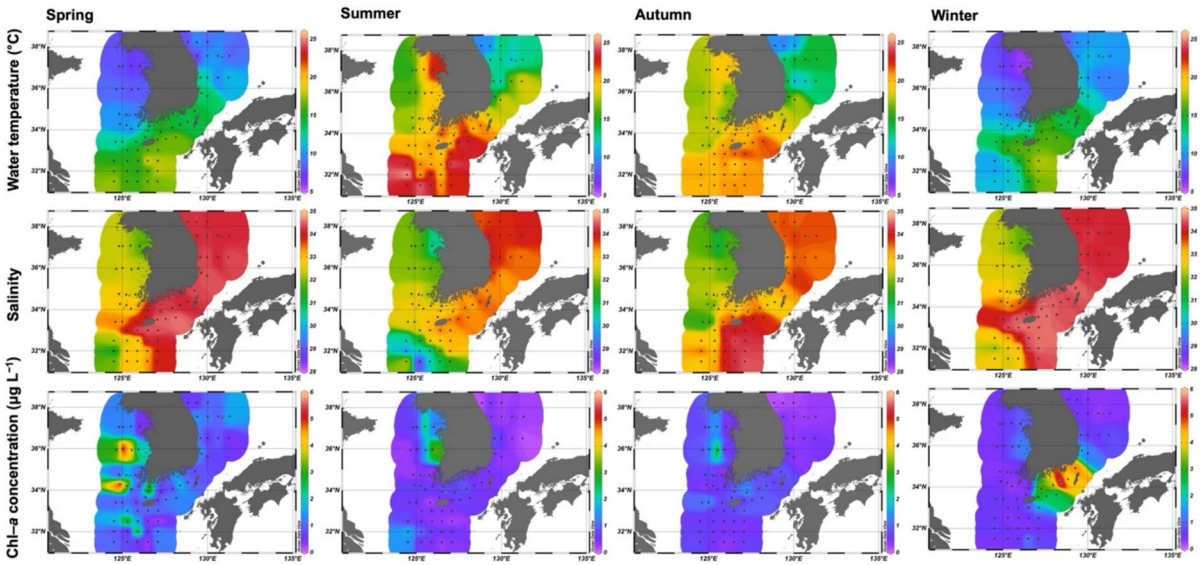

Fig. 3 Spatiotemporal variability in depth-averaged water temperature (°C), salinity, and chlorophyll-*a* (μg l^{-1}) in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS)

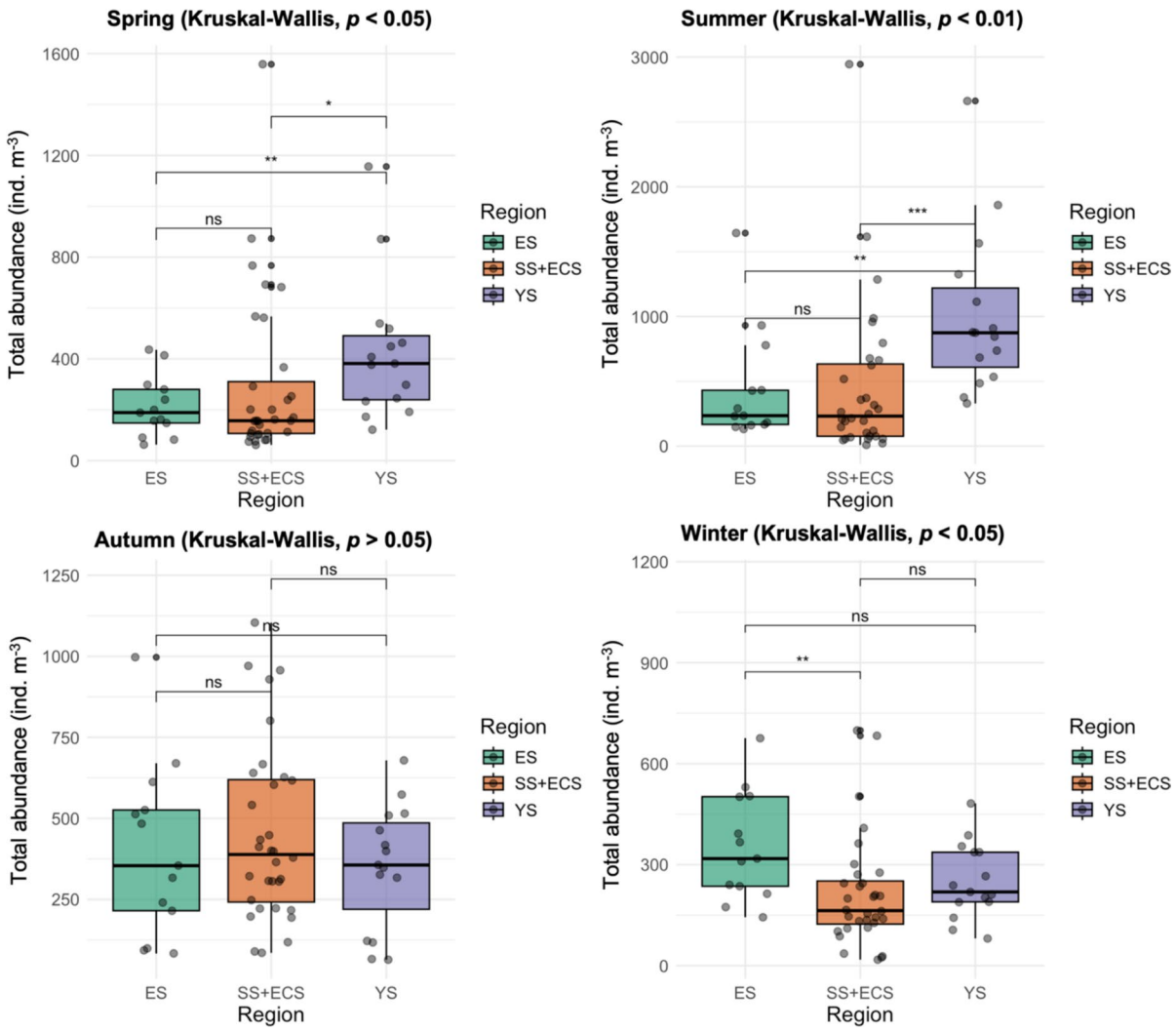

Fig. 4 Seasonal variations in copepod abundance across the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS). Box plots represent the median (horizontal line), interquartile range (box), and outliers (black dots). Kruskal–Wallis test results are shown at the top of each panel (p values). Dunn's post hoc test results indicate statistically significant differences between regions, denoted by asterisks (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). "ns" denotes non-significant differences ($p > 0.05$)

regions ($p > 0.05$). In winter, copepod abundance was highest in ES (354 ± 160 ind. m^{-3}) and lowest in SS + ECS (215 ± 166 ind. m^{-3}), with a significant difference observed between ES and SS + ECS ($p < 0.05$), while the other comparisons were not significant ($p > 0.05$). The maximum abundance in each sea area was 1,156 ind. m^{-3} at station 307–03 in YS and was particularly high at station 315–21 in SS + ECS, reaching 1558 ind. m^{-3} (Fig. 5). In summer, the average abundance was highest in the YS, followed by SS + ECS and ES. Particularly, the YS abundance was more than twice as high as that of other sea areas. The maximum abundance in each sea area was 2662 ind. m^{-3} at station 307–07 in YS, 2945 ind. m^{-3} at station 315–21 in SS + ECS, and 1643 ind. m^{-3} at station 103–05 in ES. In autumn, the average abundance was highest in SS + ECS, followed by ES and YS. The average abundance of YS decreased sharply compared to that in summer, whereas the abundance of ES and SS + ECS were similar to those observed in summer. The maximum abundance in each sea area was 679 ind. m^{-3} at station 312–03 in

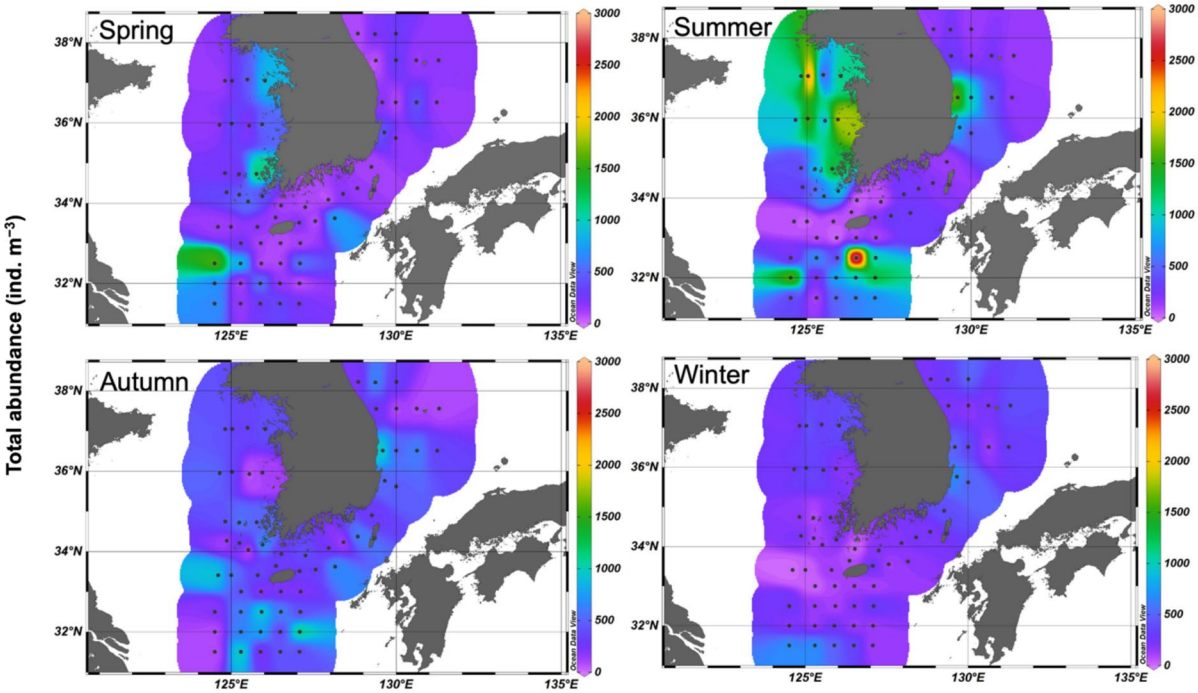

Fig. 5 Spatiotemporal variability of total abundance (ind. m^{-3}) of copepod in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS). Color gradients represent the measured values with warmer colors (red) indicating higher concentrations and cooler colors (blue) indicating lower concentrations. Black dots mark the sampling stations where data were collected during each season

YS, 1104 ind. m^{-3} at station 316–13 in SS + ECS, and 997 ind. m^{-3} at station 103–05 in ES. In winter, the average abundance was highest in the ES, followed by the YS and SS + ECS. The maximum abundance in each sea area was 482 ind. m^{-3} at station 307–07 in YS, 699 ind. m^{-3} at station 317–19 in SS + ECS, and 676 ind. m^{-3} at station 209–05 in ES.

Copepod diversity

A total of 128 species of copepods were identified throughout four seasons in the coastal waters of Korea, including the ES, SS + ECS, and YS (Supplementary Table 1). Among these, the species belonging to the order Calanoida were the most abundant, with a total of 105, followed by Cyclopoida with 23 species and a single Harpacticoida species. The seasonal changes in copepod diversity exhibited distinct patterns across the ES, SS + ECS, and the YS (Table 2). The number of species (S) was highest in the SS + ECS in autumn (24.25 ± 8.56, $p < 0.05$) and lowest in the YS in spring (8.87 ± 1.55, $p < 0.05$). Simpson's diversity index (D) peaked in the ES in autumn (7.06 ± 2.23, $p < 0.05$) and was lowest in the YS across all seasons, with the lowest values being observed in summer (2.04 ± 0.54, $p < 0.05$). The evenness index (J′), a measure of how evenly individuals are distributed across species, remained relatively stable across the regions and seasons, with slight variations. The Shannon–Weaver diversity index (H′ loge), a commonly used index accounting for both abundance and evenness of the species present in a given community, was highest in the ES in autumn (3.16 ± 0.44, $p < 0.05$) and lowest in the YS in Summer (1.94 ± 0.25, $p < 0.05$).

IndVal analysis

IndVal analysis was performed based on identified groups to assess indicator species using copepod abundance (Table 3). In spring and summer, *Oithona atlantica* (Farran, 1908), and *Metridia pacifica* (Brodsky, 1950), were consistently identified as key indicator species for Group A, while *Paraeuchaeta russelli*

Table 2 Diversity indices of copepods in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS)

Season	Index	ES	SS + ECS	YS
Spring	S	15.23 ± 3.65^{a}	12.00 ± 5.27^{a}	8.87 ± 1.55^{b}
	D	4.31 ± 1.14^{a}	3.56 ± 1.43^{a}	2.41 ± 0.43^{b}
	J′	0.94 ± 0.03^{a}	0.93 ± 0.03^{a}	0.96 ± 0.02^{a}
	H′ ($\log_e$)	2.53 ± 0.22^{a}	2.23 ± 0.39^{a}	2.08 ± 0.17^{b}
Summer	S	20.15 ± 8.51^{a}	15.53 ± 8.92^{a}	7.80 ± 2.08^{b}
	D	5.25 ± 1.95^{a}	4.24 ± 2.08^{a}	2.04 ± 0.54^{b}
	J′	0.95 ± 0.01^{a}	0.95 ± 0.03^{a}	0.96 ± 0.01^{a}
	H′ ($\log_e$)	2.78 ± 0.35^{a}	2.45 ± 0.60^{a}	1.94 ± 0.25^{b}
Autumn	S	28.54 ± 9.51^{a}	24.25 ± 8.56^{a}	10.00 ± 3.34^{b}
	D	7.06 ± 2.23^{a}	5.92 ± 1.79^{a}	2.80 ± 0.80^{b}
	J′	0.96 ± 0.02^{a}	0.96 ± 0.01^{a}	0.95 ± 0.02^{a}
	H′ ($\log_e$)	3.16 ± 0.44^{a}	3.01 ± 0.39^{a}	2.15 ± 0.29^{b}
Winter	S	21.31 ± 2.43^{a}	18.44 ± 7.76^{a}	9.33 ± 1.76^{b}
	D	5.63 ± 0.70^{a}	5.32 ± 1.98^{a}	2.64 ± 0.43^{b}
	J′	0.93 ± 0.02^{a}	0.93 ± 0.02^{a}	0.95 ± 0.01^{a}
	H′ ($\log_e$)	2.84 ± 0.11^{a}	2.61 ± 0.49^{a}	2.10 ± 0.20^{b}

S: number of species; D: Margalef's richness index; J′: Pielou's evenness index; H′ (loge): Shannon–Weaver diversity index. Values are presented as mean ± standard deviation. Different superscript letters (a, b) indicate statistically significant differences among regions within the same season ($p < 0.05$)

(Farran, 1936), and *Euchaeta plana* (Mori, 1937), were characteristic of Group C. Group B showed a predominance of *Centropages abdominalis* (Sato, 1913), in spring and *Oithona plumifera* Baird, 1843, in summer. Similar group-specific associations were found in autumn and winter, with *Canthocalanus pauper* (Giesbrecht, 1888), *Acartia negligens* (Dana, 1849), and *Microcalanus* sp. being prominent indicator species.

Threshold indicator taxa analysis (TITAN)

TITAN analysis identified seasonal thresholds along environmental gradients, revealing distinct species responses to surface temperature (STemp), bottom temperature (BTemp), surface salinity (SSali), and bottom salinity (BSali) (Figs. 6, 7, 8, 9). In spring, negative responses to lower temperature and salinity thresholds were detected in *C. abdominalis* (STemp: 11.5 °C, BSali: 34.0). Positive responses at higher thresholds occurred in *P. russelli* (BTemp: 15.4 °C, BSali: 34.5), with most positive responses found above 14 °C (water temperature) and 34.0 (salinity). In summer, positive responses at higher thresholds of bottom and surface salinity (≥ 34.0) and water temperature (≥ 26 °C) were identified in *P. russelli* (BSali: 34.25, BTemp: 12.55 °C) and *O. plumifera* (BSali: 34.15, BTemp: 6.2 °C). Negative responses below these thresholds were found in *Oithona* sp.2 (BSali: 33.8, STemp: 24.15 °C), *Paraeuchaeta elongata* (Esterly, 1913) (BTemp: 0.75 °C), and *Labidocera japonica* (Mori, 1935) (BTemp: 0.7 °C). In autumn, positive responses at higher thresholds of water temperature and salinity were observed in *A. negligens* (STemp: 21.3 °C, BSali: 34.15) and *C. pauper* (STemp: 21.5 °C, BSali: 34.35). Negative responses at lower thresholds occurred in *Mesocalanus* sp. (STemp: 19.35 °C, BTemp: 3.4 °C) and *Oithona fallax* Farran, 1913 (SSali: 31.8). In winter, positive responses at higher thresholds were found in *P. russelli* (SSali: 34.6, STemp: 15.15 °C) and *Euchaeta concinna* Dana, 1849 (SSali: 34.5, STemp: 14.55 °C), while negative responses at lower thresholds were detected in *C. abdominalis* (STemp: 10.5 °C, SSali: 32.6) and *O. fallax* (STemp: 8.85 °C, SSali: 32.6).

Seasonal occurrence of IndVal and TITAN species using Venn diagrams

The Venn diagrams illustrate the overlap in environmental factors influencing the species identified by TITAN and IndVal across four seasons (Fig. 10). In spring, six species were commonly influenced by surface temperature, bottom temperature, surface salinity, and bottom salinity, while four species were affected by a combination of bottom temperature and surface salinity. *P. elongata* and *Pseudocalanus* sp.

Table 3 Cluster-specific indicator species and their IndVal (%) list by season based on copepod abundance

Season	Group	Associations	Species	A	B	IndVal (%)	*p* value
Spring	A	ES	*Oithona atlantica*	1	1	100	0.001***
			Metridia pacifica	1	0.92	91.67	0.001***
			Clausocalanus spp.	0.9	1	90.44	0.001***
			Pseudocalanus newmani	0.96	0.83	80.4	0.001***
			Scolecithricella minor	0.96	0.83	79.81	0.001***
			Microcalanus sp.	1	0.75	75	0.001***
			Mesocalanus sp.	0.95	0.75	71.04	0.001***
			Gaetanus spp.	1	0.42	41.67	0.001***
			Paraeuchaeta elongata	1	0.33	33.33	0.002**
			Pseudocalanus sp.	1	0.33	33.33	0.001***
	B	YS	*Centropages abdominalis*	0.88	0.48	42.62	0.004**
			Oithona setigera	1	0.34	34.48	0.005**
	C	SS + ECS	*Euchaeta plana*	0.69	0.68	47.41	0.001***
			Paraeuchaeta russelli	0.94	0.47	44.46	0.001***
			Paracalanus aculeatus	0.92	0.47	43.7	0.001***
			Scolecithricella nicobarica	0.71	0.53	37.43	0.011**
			Scolecithrix danae	0.92	0.37	34.04	0.003**
Summer	A	ES	*Scolecithricella* spp.	1	1	100	0.001***
			Oithona atlantica	1	1	100	0.001***
			Metridia pacifica	1	0.92	92.31	0.001***
			Microcalanus sp.	0.98	0.85	83.18	0.001***
			Mesocalanus sp.	0.88	0.85	74.06	0.001***
			Paraeuchaeta elongata	1	0.62	61.54	0.001***
			Labidocera japonica	0.93	0.54	50.19	0.001***
			Gaetanus spp.	1	0.38	38.46	0.001***
			Pseudocalanus sp.	1	0.38	38.46	0.001***
	B	SS + ECS	*Oithona plumifera*	1	0.66	65.71	0.001***
			Paraeuchaeta russelli	0.85	0.46	38.96	0.01**
			Euchaeta plana	1	0.31	31.43	0.016*
	C	YS	*Oithona* sp.2	0.97	0.92	89.27	0.001***
			Oithona sp.1	0.89	0.33	29.69	0.005**
Autumn	A	ES	*Scolecithricella* spp.	1	0.93	92.86	0.001***
			Oithona atlantica	1	0.93	92.86	0.001***
			Metridia pacifica	1	0.86	85.71	0.001***
			Microcalanus sp.	1	0.79	78.57	0.001***
			Labidocera japonica	1	0.57	57.14	0.001***
			Mesocalanus sp.	0.91	0.57	51.96	0.001***
			Centropages furcatus	0.9	0.57	51.67	0.001***
			Paraeuchaeta elongata	1	0.5	50	0.001***
			Gaetanus spp.	1	0.43	42.86	0.001***
			Pseudocalanus newmani	1	0.43	42.86	0.001***
			Calocalanus styliremis	0.85	0.36	30.27	0.003**
			Temora turbinata	0.84	0.36	30.09	0.003**
	B	SS + ECS	*Canthocalanus pauper*	0.92	0.79	72.79	0.001***
			Oithona plumifera	1	0.58	57.58	0.001***
			Acartia negligens	1	0.36	36.36	0.003**

Table 3 (continued)

Season	Group	Associations	Species	A	B	IndVal (%)	*p* value
			Euchaeta indica	1	0.36	36.36	0.005**
			Scolecithricella longispinosa	1	0.36	36.36	0.006**
			Scolecithricella nicobarica	0.83	0.36	30.12	0.03*
	C	YS	*Oithona* sp.1	0.98	0.69	67.7	0.001***
			Oithona fallax	1	0.46	46.15	0.001***
			Acartia omorii	1	0.38	38.46	0.001***
			Centropages tenuiremis	1	0.31	30.77	0.005**
			Labidocera rotunda	1	0.31	30.77	0.002**
Winter	A	ES	*Metridia pacifica*	1	1	100	0.001***
			Scolecithricella spp.	0.97	1	97.47	0.001***
			Microcalanus sp.	1	0.92	92.31	0.001***
			Pseudocalanus newmani	1	0.85	84.62	0.001***
			Pseudocalanus sp.	1	0.85	84.62	0.001***
			Oithona atlantica	0.73	1	73.06	0.001***
			Mesocalanus sp.	0.74	0.85	62.74	0.001***
			Gaetanus spp.	1	0.62	61.54	0.001***
			Eucalanus bungii	1	0.54	53.85	0.001***
			Paracalanus gracilis	0.71	0.69	49.28	0.001***
			Acartia danae	0.79	0.46	36.56	0.003**
			Paraeuchaeta elongata	1	0.31	30.77	0.002**
	B	SS + ECS	*Paracalanus aculeatus*	0.86	1	86.4	0.001***
			Euchaeta concinna	0.94	0.88	82.3	0.001***
			Euchaeta plana	0.82	0.88	71.99	0.001***
			Clausocalanus farrani	0.83	0.75	62.35	0.001***
			Candacia bipinnata	0.84	0.63	52.75	0.001***
			Oithona plumifera	0.9	0.56	50.81	0.001***
			Paraeuchaeta russelli	0.8	0.63	50.23	0.001***
			Cosmocalanus darwinii	0.95	0.5	47.65	0.001***
			Oithona spp.	0.9	0.5	45.09	0.001***
			Scolecithricella longispinosa	0.65	0.69	44.55	0.015*
			Subeucalanus subtenuis	0.95	0.44	41.78	0.001***
			Pareucalanus attenuatus	1	0.31	31.25	0.001***
			Clausocalanus furcatus	0.98	0.31	30.61	0.002**
	C	YS	*Centropages abdominalis*	0.91	0.32	29.2	0.02*
			Oithona fallax	1	0.29	29.03	0.015*

Asterisks indicate significant differences (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). Site clusters were identified as follows: Group A = East Sea (ES), Group B = Yellow Sea (YS), and Group C = South Sea + East China Sea (SS + ECS)

were exclusively identified as indicator species by IndVal. In summer, five species were influenced by the intersection of all environmental variables, while four species were uniquely responsive to bottom temperature. In autumn, seven species were impacted by the overlap of all four variables, while four species responded solely to bottom temperature. IndVal identified *Calocalanus styliremis* (Giesbrecht, 1888), *Centropages tenuiremis* (Thompson I.C. & Scott A., 1903), *Labidocera rotunda* (Mori, 1929), *Scolecithricella nicobarica* (Sewell, 1929), and *Temora turbinata* (Dana, 1849), as indicator species that were not identified in TITAN. In winter, 19 species were commonly influenced by all four environmental variables,

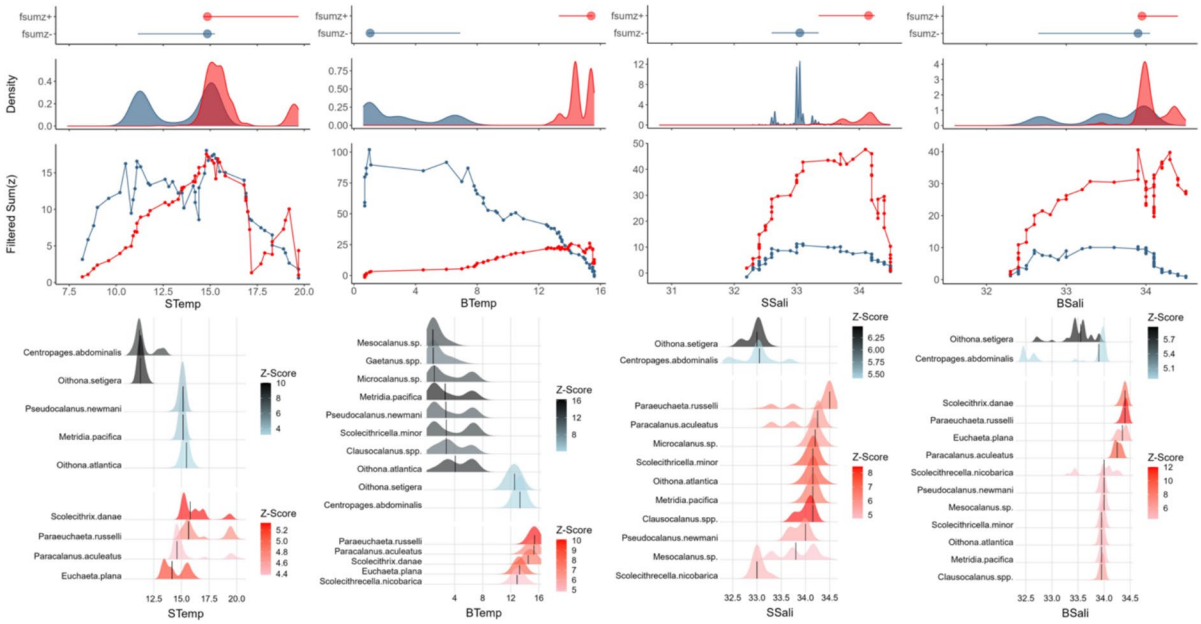

Fig. 6 TITAN (Threshold Indicator Taxa Analysis) results illustrating copepod species responses to environmental gradients in spring. The four panels represent responses to surface temperature (STemp), bottom temperature (BTemp), surface salinity (SSali), and bottom salinity (BSali). The top row shows threshold responses, where red dots indicate species exhibiting positive responses (sum [z^+]) and blue dots represent species showing negative responses (sum [z^-]). The density plots highlight the distribution of threshold change points (CP) for increasing and decreasing trends. The middle row presents filtered sum (z) scores across the environmental gradient, visualizing species' aggregated response patterns. The bottom row displays individual species responses, with dark blue indicating negative responses (z^-) and red representing positive responses (z^+). The Z-score intensity reflects the magnitude of the species' response to environmental thresholds

with three species responding specifically to bottom temperature. *P. elongata* was solely identified by IndVal, while *P. attenuatus* was found at the intersection of surface temperature and bottom salinity.

Functional traits of indicator species

The functional trait matrix based on indicator species is presented in Supplementary Table 2. The functional traits of indicator species extracted based on clusters varied temporally and spatially (Fig. 11). In the YS, broadcasters, 1–2 mm length species, omnivores, and mixed feeders were predominantly observed during spring. In summer, sac-spawners, < 1 mm length species, omnivores, and ambush feeders became dominant. In autumn, broadcasters and sac-spawners were observed in similar proportions, whereas in winter, sac-spawners, < 1 mm length species, omnivores, and ambush feeders were more prevalent. However, during autumn, only 2–3 mm, O–C, and current feeding species were observed. The functional traits in the YS exhibited following trends: the spawning strategy exhibited a decrease in the proportion of broadcasters over time, with an increase in sac-spawners; body length shifted toward smaller species (< 1 mm) over time. In terms of trophic groups, omnivores were generally predominant, and although mixed feeders were prevalent in certain seasons, ambush feeders were also commonly observed.

In SS + ECS, during spring, broadcasters and sac-spawners occurred in similar proportions and were composed of > 3 mm species encompassing all five trophic groups (omnivore, carnivore, O–C, O–D, and O–H) and 3 feeding types (ambush, mixed, and NA). In summer, sac-spawners, 1–2 mm, and > 3 mm species were present in similar proportions, primarily consisting of the carnivore group (carnivore, O–C) and encompassing three feeding types (ambush, current, mixed). During autumn, the community composition mainly included sac-spawners, 1–2 mm species, three trophic groups (O–C, O–D, and O–H), and three feeding types (current, ambush, mixed, in that order).

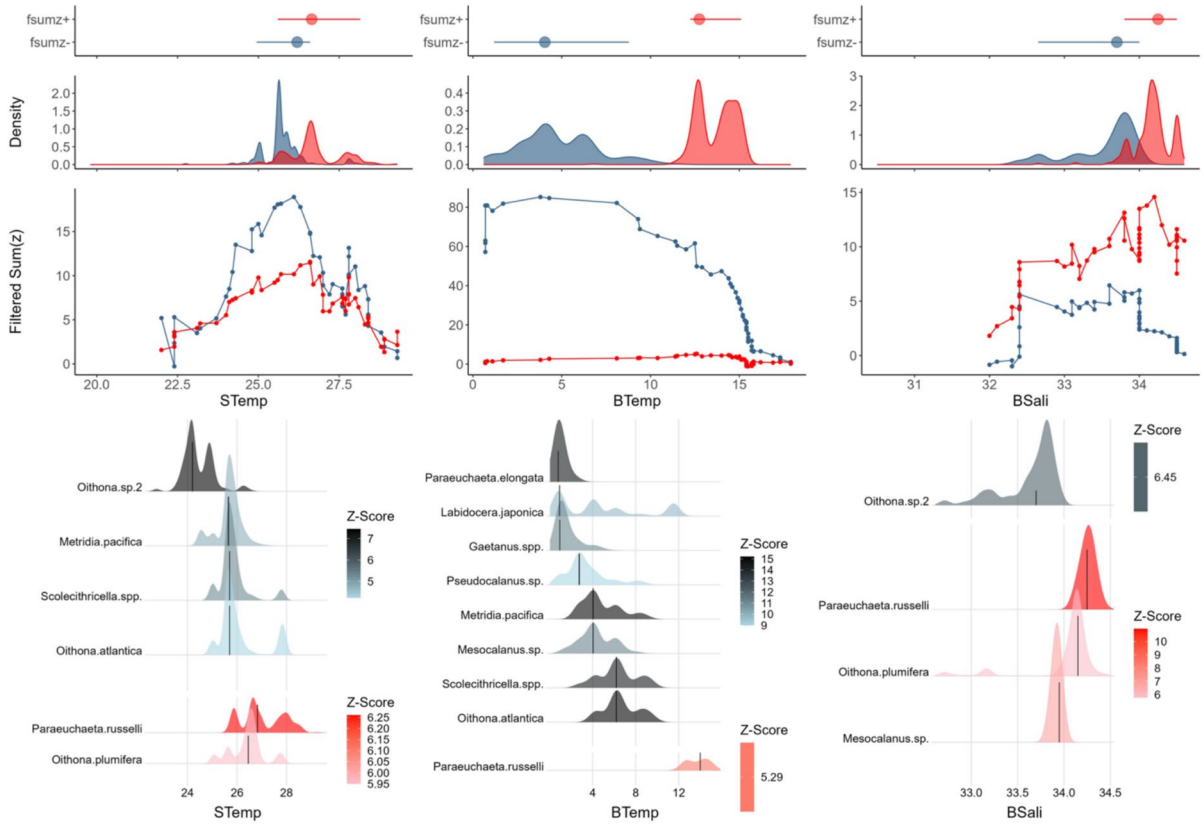

Fig. 7 TITAN (Threshold Indicator Taxa Analysis) results illustrating copepod species responses to environmental gradients in summer. The four panels represent responses to surface temperature (STemp), bottom temperature (BTemp), and bottom salinity (BSali). The top row shows threshold responses, where red dots indicate species exhibiting positive responses (sum [z^+]) and blue dots represent species showing negative responses (sum [z^-]). The density plots highlight the distribution of threshold change points (CP) for increasing and decreasing trends. The middle row presents filtered sum (z) scores across the environmental gradient, visualizing species' aggregated response patterns. The bottom row displays individual species responses, with dark blue indicating negative responses (z^-) and red representing positive responses (z^+). The Z-score intensity reflects the magnitude of the species' response to environmental thresholds

The copepod community in winter was composed of sac-spawners, 1–2 mm species, five trophic groups (omnivore, carnivore, O–C, O–D, and O–H), and four feeding types (ambush, current, cruise, NA, in that order). The functional traits in SS + ECS exhibited similar trends to those of the YS, with a decrease in broadcasters and an increase in sac-spawners over time. Body length shifted toward a dominance of 1–2 mm species over time. Trophic groups exhibited a shift from a carnivore dominance to a decrease in O–H during autumn and winter. Regarding feeding types, the communities mainly featured ambush feeders followed closely by current feeders, except during summer when only ambush feeders were observed.

In the ES, sac-spawners, 1–2 mm species, O–H, and cruise feeders were predominant during spring. In summer, broadcasters, 1–2 mm, omnivores, and current feeders were dominant. In autumn, broadcasters, 1–2 mm species, omnivores, and current feeders once again became prevalent. In winter, broadcasters, 1–2 mm, and 2–3 mm species were present in similar proportions, with omnivores and current feeders being dominant. The functional traits in the ES exhibited the following trends: contrary to the YS and SS + ECS, the spawning strategy showed a decrease in the proportion of sac-spawners and an increase in broadcasters. Body length exhibited a decrease in 1–2 mm species and an increase in 2–3 mm species over time. Trophic groups showed a decrease in O–H

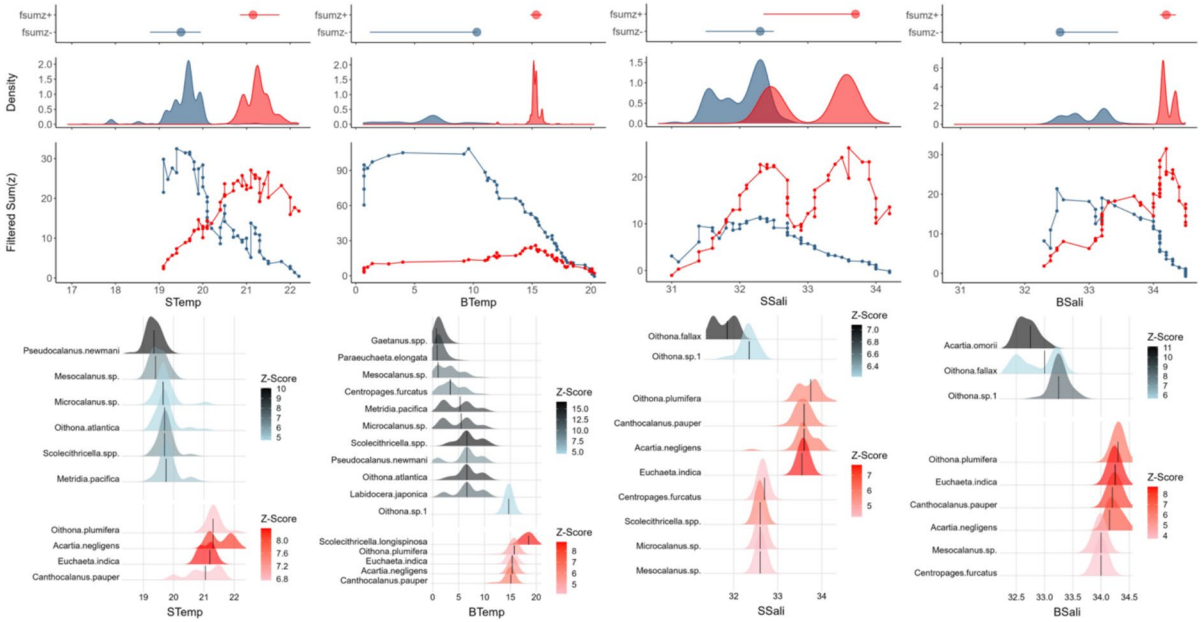

Fig. 8 TITAN (Threshold Indicator Taxa Analysis) results illustrating copepod species responses to environmental gradients in autumn. The four panels represent responses to surface temperature (STemp), bottom temperature (BTemp), and bottom salinity (BSali). The top row shows threshold responses, where red dots indicate species exhibiting positive responses (sum [z^+]) and blue dots represent species showing negative responses (sum [z^-]). The density plots highlight the distribution of threshold change points (CP) for increasing and decreasing trends. The middle row presents filtered sum (z) scores across the environmental gradient, visualizing species' aggregated response patterns. The bottom row displays individual species responses, with dark blue indicating negative responses (z^-) and red representing positive responses (z^+). The Z-score intensity reflects the magnitude of the species' response to environmental thresholds

throughout the seasons, with omnivores replacing them. Regarding feeding types, there was a decrease in cruise feeders, whereas current feeders and ambush feeders became more prevalent, gradually taking over that ecological niche.

Distance-based redundancy analysis (dbRDA) of indicator species

The dbRDA analysis using environmental factors and indicator species data was conducted with surface and bottom data for temperature, salinity, and Chl-*a* (Fig. 12, Table 4). The results indicate that the environmental variables used for prediction effectively explained ECS + SS, ES, and YS. In spring, the first and second dbRDA axes explained 49.9% of the variation, with surface and bottom temperature, as well as bottom salinity, showing statistically significant relationships with the axes ($p < 0.001$). SS + ECS exhibited large variation along the first axis, with surface and bottom temperature being the major contributing factors, showing a positive correlation. In contrast, ES showed smaller variation along both axes and had a negative correlation with bottom temperature. YS displayed a negative correlation with surface temperature but a weak positive correlation with surface and bottom Chl-*a*. In summer, the first and second dbRDA axes explained 62.8% of the variation, with temperature and salinity distinguishing the regions more clearly than Chl-*a* ($p < 0.001$). SS + ECS showed a strong positive correlation with bottom salinity and surface temperature, with less variation compared to spring. ES exhibited a negative correlation with bottom temperature and Chl-*a*, similar to spring, while showing a weak positive correlation with surface salinity. YS had a negative correlation with bottom salinity and surface temperature but a weak positive correlation with Chl-*a*. In autumn, the first and second dbRDA axes explained 52.8% of the variation, with all environmental variables showing statistically significant contributions ($p < 0.05$). Notably, SS + ECS had strong positive correlations

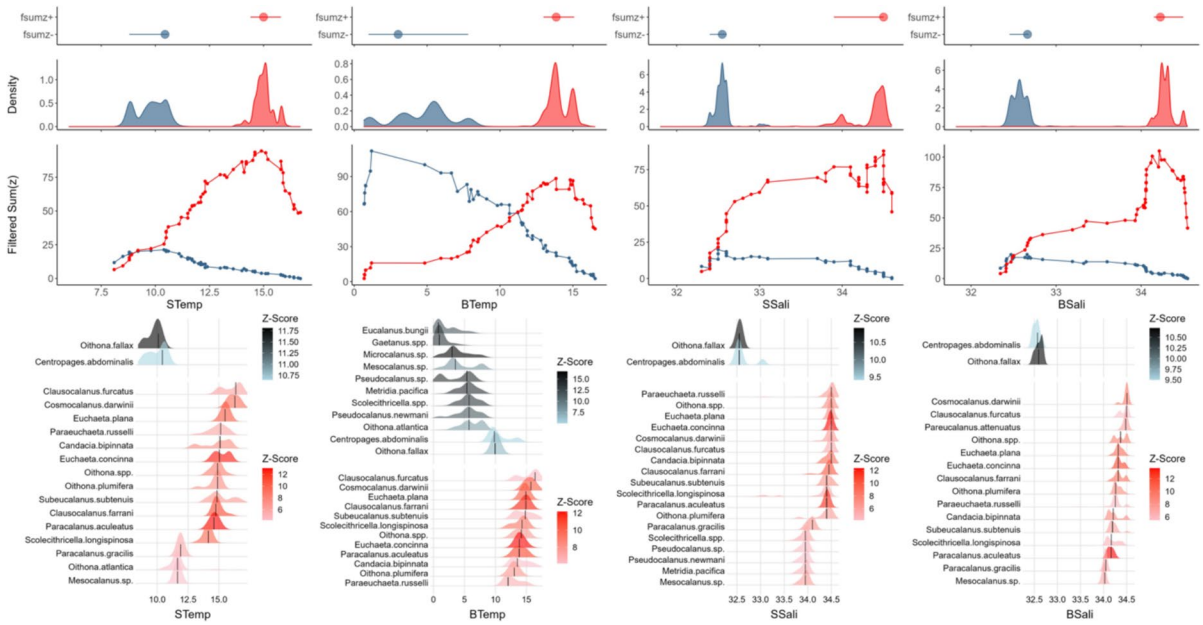

Fig. 9 TITAN (Threshold Indicator Taxa Analysis) results illustrating copepod species responses to environmental gradients in winter. The four panels represent responses to surface temperature (STemp), bottom temperature (BTemp), and bottom salinity (BSali). The top row shows threshold responses, where red dots indicate species exhibiting positive responses (sum [z^+]) and blue dots represent species showing negative responses (sum [z^-]). The density plots highlight the distribution of threshold change points (CP) for increasing and decreasing trends. The middle row presents filtered sum (z) scores across the environmental gradient, visualizing species' aggregated response patterns. The bottom row displays individual species responses, with dark blue indicating negative responses (z^-) and red representing positive responses (z^+). The Z-score intensity reflects the magnitude of the species' response to environmental thresholds

with surface and bottom temperature and salinity. ES showed a negative correlation with temperature rather than salinity. YS exhibited a negative correlation with salinity and a weak negative correlation with bottom Chl-*a*. In winter, the first and second axes of dbRDA explained 48.0% of the variation, and the results indicated that all environmental factors, except for bottom Chl-*a*, had a significant influence ($p < 0.05$). SS + ECS showed positive correlations with temperature and salinity, whereas YS exhibited negative correlations with these variables. Some stations showed an overlap between YS and SS + ECS, but most stations remained well separated. ES displayed a negative correlation with bottom temperature and a positive correlation with salinity.

Discussion

This study provides a detailed examination of copepod diversity and functional traits across the marine ecosystems surrounding the Korean Peninsula, including the ES, SS, YS, and ECS. By evaluating the observed spatial and temporal variations, we have gained insights into the potential ecological roles of copepods in diverse marine environments. These results underline the significance of copepods in sustaining ecosystem resilience and trophic interactions in these marine ecosystems.

Hydrographic conditions

Our study revealed important hydrographic changes affecting the distribution and seasonal abundance of copepods across marine environments. The observed seasonal fluctuations in water temperatures, particularly the elevated temperatures in the SS + ECS during summer, suggest an influence of ocean currents. The ECS, located on the northwestern continental shelf of the Pacific Ocean, is predominantly influenced by the CDW, the TWC, the KC, and the YSCWM (Chen & Wang, 1999; Lie & Cho, 2016; Liu et al., 2021).

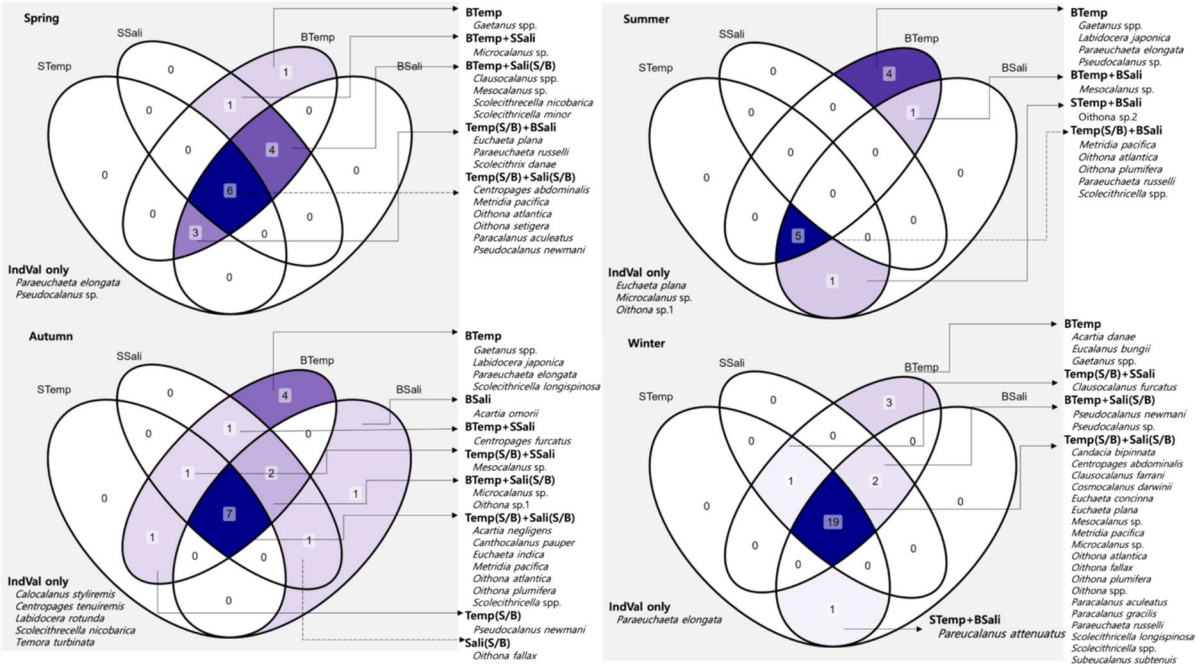

Fig. 10 Venn diagrams illustrating the overlap in environmental factors influencing the species identified by TITAN and IndVal in each season. Each diagram represents the influence of surface temperature (STemp), bottom temperature (BTemp), surface salinity (SSali), and bottom salinity (BSali). The numbers within each section indicate the number of species responding to the corresponding environmental variable(s). The dark blue regions show species responding to all four variables, while the lighter shaded areas represent species affected by subsets of these variables. Species listed in the IndVal only section were exclusively identified through IndVal analysis and were not detected by TITAN

The mixing of these water masses varies depending on the season, leading to diverse hydrographic conditions (Ichikawa & Beardsley 2002). Notably, during the rainy summer season, increased discharge from the Yangtze River leads to the expansion of the CDW, characterized by salinity levels below 31.0, extending to areas as far as Jeju Island (Isobe et al. 2004). In our study, the SS + ECS area, specifically along lines 315, 316, and 317, exhibited salinity levels below 31 at seven stations. Changes in water temperature and salinity are significant factors that influence the composition and distribution of zooplankton, as these environmental variables directly affect biological processes such as reproduction, survival, and migration patterns of copepod species (Menéndez et al., 2016; Choi et al., 2021b). Our findings are consistent with those of prior studies highlighting the pivotal role of hydrographic conditions (e.g., seasonal fluctuations in water temperature and salinity) in shaping marine biodiversity (Li et al., 2022; Shin et al., 2022; Choo et al., 2023).

Sustained climate change is expected to further alter these hydrographic conditions, exerting distinct impacts on copepod assemblages. Climate-induced warming strengthens ocean stratification, modifying nutrient fluxes and ultimately affecting primary productivity, which serves as a fundamental energy source for copepod populations (Suikkanen et al., 2013). Seasonal and warming-induced changes may disrupt the timing of phytoplankton blooms, leading to mismatches between copepod spawning cycles and food availability, ultimately influencing recruitment success and population stability (Edwards & Richardson, 2004). These shifts may favor smaller-bodied, warm-adapted copepod species while leading to declines in cold water taxa, reflecting broader trends in marine ecosystem transformations (Richardson, 2008; Batten & Walne, 2011). Functionally, this may result in the decline of lipid-rich copepods such as *Calanus finmarchicus* (Miller et al., 2000), which play a crucial role in energy transfer to higher trophic levels while increasing the dominance of opportunistic,

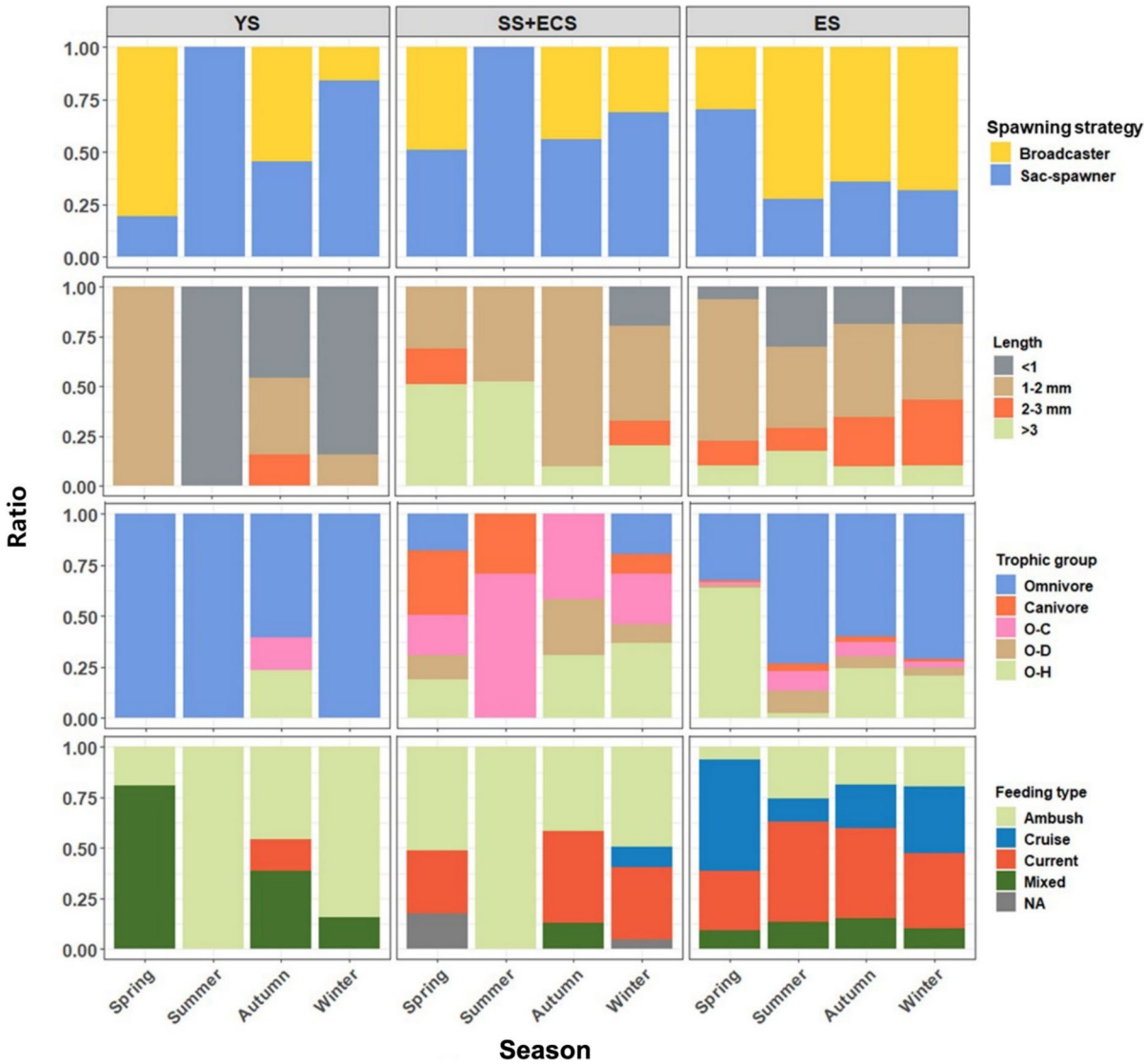

Fig. 11 Seasonal distributions and variations of each functional trait in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS)

omnivorous taxa (Protopapa et al., 2019). Furthermore, increased anthropogenic impacts, such as coastal eutrophication and pollution, may exacerbate these changes by altering phytoplankton composition and trophic interactions, further influencing copepod community dynamics (Roman et al., 2019; Pierson et al., 2020). Beyond seasonal variability, interannual processes such as ENSO and PDO also influence copepod assemblages by modifying oceanographic conditions over multi-year periods (Beaufort & Grelaud, 2017). ENSO–driven shifts in upwelling and nutrient fluxes can lead to long-term changes in copepod abundance and species composition (Chavez et al., 2003; Hooff & Peterson, 2006; Beaufort & Grelaud, 2017), highlighting the need for extended monitoring to capture both short-term fluctuations and long-term ecosystem trends.

Assemblage structure and dynamics of copepods

The results from our study provide a comprehensive overview of the copepod distribution and community

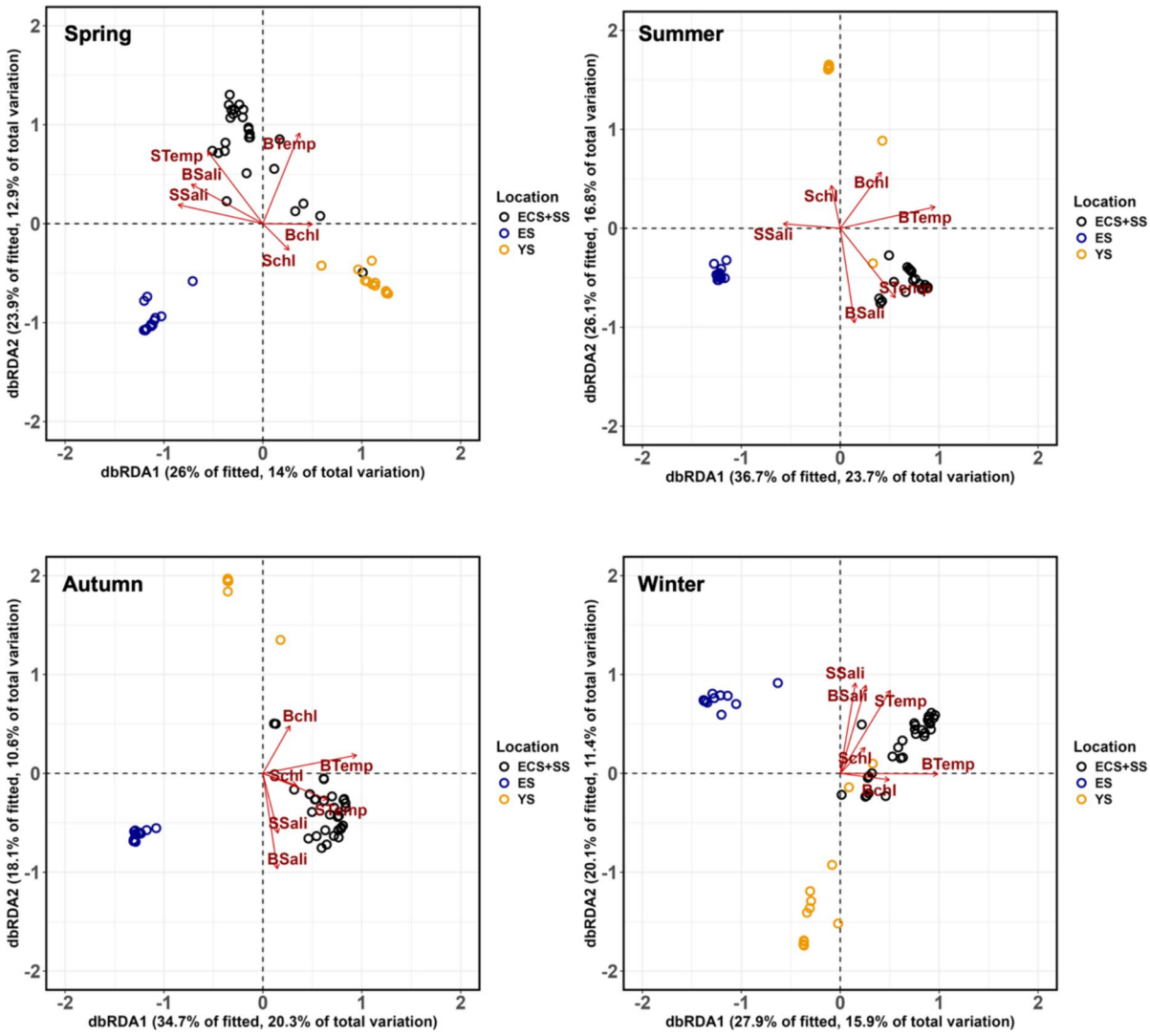

Fig. 12 Distance-based Redundancy Analysis (dbRDA) of copepod communities in relation to environmental variables across seasons. Each plot represents a different season (spring, summer, autumn, and winter), showing the relationships between environmental variables and Indval species at different locations in the East Sea (ES), South Sea + East China Sea (SS + ECS), and Yellow Sea (YS). Environmental variables are represented as red biplot arrows, including surface temperature (STemp), bottom temperature (BTemp), surface salinity (SSal), bottom salinity (BSal), surface Chl-*a* (Schl), and bottom Chl-*a* (Bchl)

structure within the ES, SS + ECS, and YS. These findings revealed distinct copepod assemblages across three marine regions in a season-dependent manner, thus demonstrating the clear spatiotemporal variation of copepod communities in the study area. In the ES, the observed peak in copepod diversity during autumn can be directly linked to the significant environmental shifts that characterize this season. Kwon et al. (2019) reported that autumnal changes in the ES, including variations in nutrient salts and the weakening of the thermocline, lead to higher nutrient concentrations. These conditions are further enhanced by the influx of external water masses and coastal upwelling, which collectively contribute to an enriched environment supportive of a diverse copepod community. Such an influx of nutrient-rich waters in autumn creates the optimal conditions for a wide array of copepod species, fostering a period of

Table 4 Results of the distance-based redundancy analysis (dbRDA) showing the influence of environmental variables on different layers across seasons

Season	Layer	Environmental variables	df	SumSqs	F	*p* value
Spring	Surface	Water temperature (°C)	1	4.1547	17.6844	**0.001**
		Salinity	1	2.8121	11.9697	**0.001**
		Chl-*a* (μg l^{-1})	1	0.2871	1.222	0.284
	Bottom	Water temperature (°C)	1	2.9354	12.4945	**0.001**
		Salinity	1	0.3246	1.3815	0.2
		Chl-*a* (μg l^{-1})	1	0.2434	1.0361	0.389
Residual			46	10.8072		
Summer	Surface	Water temperature (°C)	1	3.9188	22.6594	**0.001**
		Salinity	1	1.2566	7.2657	**0.001**
		Chl-*a* (μg l^{-1})	1	0.1252	0.7242	0.543
	Bottom	Water temperature (°C)	1	3.9097	22.6072	**0.001**
		Salinity	1	1.4406	8.3297	**0.001**
		Chl-*a* (μg l^{-1})	1	0.0542	0.3132	0.896
Residual			42	7.2636		
Autumn	Surface	Water temperature (°C)	1	2.9933	11.8751	**0.001**
		Salinity	1	1.5992	6.3446	**0.001**
		Chl-*a* (μg l^{-1})	1	0.8288	3.2881	**0.004**
	Bottom	Water temperature (°C)	1	3.4327	13.6183	**0.001**
		Salinity	1	1.5391	6.1058	**0.001**
		Chl-*a* (μg l^{-1})	1	0.5289	2.0981	**0.029**
Residual			51	12.8553		
Winter	Surface	Water temperature (°C)	1	4.4209	19.0621	**0.001**
		Salinity	1	2.789	12.0257	**0.001**
		Chl-*a* (μg l^{-1})	1	0.9724	4.193	**0.001**
	Bottom	Water temperature (°C)	1	2.5223	10.8756	**0.001**
		Salinity	1	0.6837	2.948	**0.006**
		Chl-*a* (μg l^{-1})	1	0.1964	0.8468	0.565
Residual			52	12.0599		

Significant effects ($p < 0.05$ and $p < 0.001$) are highlighted in bold

seasonal richness that is distinct from other times of the year (Lee & Lee, 2017). Such seasonal richness in the ES contrasts with the more uniform distribution seen in the YS, where the summer season exhibits a spike in copepod abundance but not necessarily in diversity. This limited increase in species diversity suggests that these specific environmental conditions promote the proliferation of only a select group of copepod species, potentially limiting the overall biodiversity within the YS during this period (Wang et al., 2003; Zhang et al., 2008).

The SS + ECS regions are particularly known for their high copepod diversity and abundance, especially in summer and autumn, which can be attributed to the complex interplay of the KC, monsoonal effects, and the influence of freshwater discharges, which significantly influence salinity and nutrient dynamics (Shin et al., 2022; Choo et al., 2023). The interplay of these factors maintains conditions conducive to phytoplankton growth, which is pivotal for sustaining high zooplankton and fish productivity in these marine ecosystems (Chen et al., 2017; Park et al., 2022). The resulting increase in primary productivity, driven by warm currents that mix with cooler, nutrient-rich waters, underpins the rich diversity and abundance of copepod species in the SS + ECS.

The integration of TITAN and IndVal analyses using Venn diagrams provided key insights into the environmental factors influencing copepod communities across different seasons (Supplementary Table 3). The TITAN analysis detected species with threshold responses to environmental variability, particularly bottom temperature and salinity, whereas

IndVal identified species characteristic of specific regions and seasonal conditions. The seasonal shifts in shared species indicated that temperature-sensitive species dominated in spring and summer, reflecting the effects of seasonal warming and stratification. In contrast, autumn and winter were characterized by species associated with salinity gradients, suggesting that freshwater input and vertical mixing played a more significant role in shaping community composition. The notably high number of species responding to both temperature and salinity in winter suggests increased vertical mixing and reduced stratification, leading to a more homogeneous hydrographic environment. The integration of TITAN and IndVal enhances the comprehensive assessment of copepod responses, capturing both threshold-driven species and spatially characteristic taxa, thereby reinforcing their utility as bioindicators for monitoring environmental changes in marine ecosystems.

Functional traits of indicator species

Functional traits are phenotypic characteristics of organisms that are related to ecological functions (Violle et al., 2007; Benedetti et al., 2016). Given that organisms must allocate energy to perform their ecological functions, functional traits are closely linked through trade-off relationships (Kiørboe, 2011; Litchman et al., 2013). Classification based on functional traits has been used across a broad range of ecosystems, from terrestrial to marine ecosystems (Fu et al., 2014; Moretti et al., 2017; Harvey et al., 2022; Boutin et al., 2023). A comprehensive matrix for the major functional traits of zooplankton has only recently been established. However, despite significant efforts, the number of species included in this matrix remains limited, posing challenges due to insufficient information for applications in research related to marine ecosystems (Litchman et al., 2013; Benedetti et al., 2016). This lack of information poses a challenge for future research on marine ecosystem dynamics. Despite these shortcomings, functional traits allow for species with similar characteristics to be grouped according to their ecosystem functions and environmental conditions, thus providing insights into the functional diversity and ecological niches within a given space, as well as intra- and/or inter-group competition (Gitay & Noble, 1997; Vogt et al., 2013; Pomerleau et al., 2015; Benedetti et al., 2018).

In this study, we conducted cluster analysis on the most abundant copepod species collected in the study area, focusing on the zooplankton, to identify species that could serve as indicators to distinguish between seasons and regions. The functional traits of these indicator species effectively reflected the seasonal and spatial characteristics of their respective regions, with statistical validation provided by TITAN, dbRDA, and Venn diagram. The YS was primarily influenced by seasonal fluctuations in the YSCWM, whereas the SS + ECS were influenced by factors such as the CDW, the KC, and the TWC, and the ES exhibited influences from the TWC and the North Korean Cold Current (NKCC). The composition of species and their functional traits varied according to the environmental characteristics of distinct regions. Given the challenges in explaining the state of a region based on individual traits of diverse copepod species, it is common to group them based on shared functions and describe the characteristics of functional groups to elucidate the overall state of the region. However, this is the first study to interpret the functional ecology of copepods by integrating the areas surrounding South Korea, including the YS, SS + ECS, and ES.

In the YS, the functional traits of indicator species exhibited similarities between summer and winter and between spring and autumn. Generally, the indicator species within copepod communities in the YS were predominantly omnivore (Fig. 11). The main shared traits during summer and winter included sac-spawners, small copepods of 1 mm or less, and ambush feeding types. TITAN results confirmed that abrupt shifts in species abundance occurred in response to temperature and salinity changes, indicating species-specific thresholds for environmental stress. Notably, species such as *C. abdominalis* and *O. setigera* showed strong negative responses to decreasing bottom temperature and surface salinity, with threshold values around 13 °C and 33, respectively. The dbRDA results confirmed that YS copepod assemblages were mainly influenced by bottom salinity and bottom temperature, with variations in seasonal importance. In spring, dbRDA revealed that surface and bottom temperature, as well as bottom salinity ($p < 0.001$), were significant predictors, with YS assemblages exhibiting a weak negative correlation with surface temperature but a positive correlation with bottom Chl-*a*. This suggests that food availability is a key driver of copepod distributions during this season. In contrast,

during winter, bottom salinity and temperature were the dominant environmental drivers, while autumn patterns showed a stronger influence from bottom chlorophyll, aligning with increased primary production. Overall, the combined results from dbRDA and TITAN emphasize the pivotal role of bottom salinity and temperature in structuring YS copepod communities. The overlap between methods highlights species such as *C. abdominalis* and *O. setigera* as key indicators of hydrographic variability, reflecting their adaptation to dynamic environmental conditions and their potential utility as bioindicators of marine ecosystem shifts. In contrast, the dominant shared traits during spring and autumn included broadcaster and mixed feeding type species. However, during autumn, there was an observed increase in the proportion of larger-sized groups engaged in current feeding compared to spring. The YS is known for the presence of cold water in the bottom layer throughout the year (Yang et al., 2014; Li et al., 2016). In summer, there is a significant temperature difference between the surface layer and the bottom layer due to convective mixing, with limited food supply in some surface mixing layers (Ren et al., 2014; Kim et al., 2018). In winter, especially in the relatively shallow western part, temperature decreases from the surface to the bottom layers, resulting in a drastic decrease in food sources due to the impact of cold water from the YSCWM (Yin et al., 2013). In such an environment, the hatching time of egg sac copepods may become delayed (Choi et al., 2021a), and the increase in ambush feeding can be interpreted as a strategy to conserve energy for foraging (Litchman et al., 2013; Li et al., 2022). The trends observed in spring and autumn, as supported by the dbRDA results, suggest an improvement in the food environment compared to winter and summer. Particularly, a relatively higher proportion of larger-sized groups was observed in spring compared to indicator species in summer or winter, along with an increase in the proportion of broadcasters (Fig. 11). This suggests that seasonal shifts in functional traits are closely linked to hydrographic changes, with copepods in the YS adapting to dynamic environmental conditions through shifts in reproductive strategies and feeding behaviors. Regarding feeding type, the adoption of mixed feeding marks a shift in functional strategy toward more active feeding behavior, thus establishing new feeding patterns that ensure the preservation of future populations (Jansson et al., 2020).

Except for summer, SS + ECS exhibited similar traits throughout all seasons (Fig. 11). Regarding the seasonal similarities related to functional traits, the proportion of broadcasters and sac-spawners remained at approximately 50%, and the proportion of ambush and current feeders was also approximately 50%. dbRDA analysis revealed that the SS + ECS regions were influenced by the KC and TWC year-round, with the inflow of the CDW during summer further affecting salinity and nutrient levels (Lie & Cho, 2016). TITAN results confirmed that abrupt shifts in species abundance occurred in response to bottom salinity and temperature fluctuations, with species-specific thresholds. Notably, species such as *E. plana* and *P. russelli* showed strong responses to increasing temperature and salinity, indicating their adaptability to changing hydrographic conditions. The dbRDA results revealed that bottom temperature and surface temperature were the primary environmental drivers influencing SS + ECS copepod assemblages, with variations in seasonal importance. In spring, dbRDA indicated that surface and bottom temperature were significant predictors, with SS + ECS copepod communities exhibiting a strong correlation with increasing temperature. In summer, bottom salinity and surface temperature played dominant roles, while in autumn, ECS + SS exhibited strong positive correlations with surface and bottom temperature and salinity. In winter, temperature and salinity continued to structure the community, but the influence of surface Chl-*a* also increased, suggesting seasonal variability in hydrographic drivers. Therefore, the indicator copepods within the SS + ECS were generally composed of species originating from foreign seas (Shin et al., 2022; Choo et al., 2023). The waters of the SS + ECS are known to have a better food environment throughout the year than other waters due to the formation of a thermohaline front and salinity front (Morgan et al., 2005). Additionally, the species originating from foreign seas introduced by ocean currents in these environments are generally larger in size and are characteristically more diverse. These copepod species occurring in the SS + ECS tend to exhibit current feeding throughout the year except for summer and display various feeding-associated functional traits such as

carnivorous (or omnivore–carnivore) and omnivorous-detritivorous (or omnivore) behaviors. However, the influence of environmental factors cannot be overlooked, as 2020 witnessed record high rainfall levels, and river discharge was also higher than usual during that period (Cui et al., 2022). The dbRDA also showed that record high rainfall and increased river discharge in 2020 led to salinity-related stress responses, influencing the species composition during that year. The occurrence of salinity-related stress responses and/or the transport of estuarine–marine copepods also suggest an increase in organic matter, including carcasses, in a food-rich environment dominated by phytoplankton (Jeong et al. 2022). This rapid change in the environmental conditions explains the increased proportion of ambush feeding copepod species, along with the response of strategic sac-spawners and an increase in the proportion of carnivore and omnivore–carnivore species.

Similar seasonal traits were observed in the ES (Fig. 11). The deep-sea regions of the ES, being relatively deep, maintain a stable ecosystem that preserves its unique features, rather than significantly responding to seasonal variations (Lee et al., 2009). The dbRDA findings emphasized that, despite persistent food scarcity, the ES had relatively stable temperature and salinity conditions (Lee et al., 2017), which were reflected in the consistent proportions of broadcaster and sac-spawner species throughout the year, as well as the consistent variation in size groups, except for spring. Encounters with prey can be passively and directly blocked by protozoans, and this prey type influences prey selection (Kiørboe, 2011). For example, ambush feeding enables low energy expenditure and low predator encounter rates but results in relatively low prey encounter efficiency and mating encounter rates (Kiørboe et al., 2010). However, mixed feeders can strike a balance between proper feeding efficiency and the ability to maximize growth and survive in low prey conditions. In these contexts, except for spring, the dominance of omnivores and the combination of current + cruise feeding types throughout the year emphasize the need to allocate more energy toward actively searching for food. In the ES, the increase in size groups corresponding to the 2–3 mm range over time can be interpreted as the maintenance of similar traits rather than the influx of new functional traits.

Conclusion

This study provides a comprehensive analysis of copepod communities in the ES, YS, and the combined SS + ECS, emphasizing their functional traits as ecological indicators. Variations in the hydrological and environmental conditions unique to each region lead to spatial and temporal variations in copepod community structure. To summarize our results: (i) The YS copepod communities demonstrated significant seasonal shifts, with small omnivorous species dominating in spring and winter, while sac-spawners became prevalent during summer. The SS + ECS region, influenced by the KC and the CDW, exhibited higher species richness and functional diversity throughout the year, with ambush feeders being particularly dominant in nutrient-rich conditions. The ES, characterized by its stable deep-sea conditions, showed less pronounced seasonal variation but a steady dominance of sac-spawners and larger omnivores. (ii) Environmental factors such as temperature, salinity, and Chl-*a* concentrations were identified as primary influences on the observed community structures, with the YS's cold water masses and the SS + ECS's complex currents significantly shaping seasonal community variations. (iii) Copepod functional traits, including feeding strategies, body size, and reproductive methods, are effective indicators of environmental conditions, offering valuable insights into marine ecosystem health. While this study offers valuable insights into copepod diversity and functional ecology, the findings are derived from observations conducted over a single year. Consequently, extended long-term surveys are necessary to fully capture temporal variations and ensure a more comprehensive understanding.

Author contributions Seo Yeol Choi contributed toward methodology, software, formal analysis, data curation, and writing—original draft. Ho Young Soh contributed toward methodology, conceptualization, and review & editing. Min Ho Seo contributed toward investigation and methodology. Eun Hye Lee contributed toward methodology and conceptualization. Young Seok Jeong contributed toward methodology, software, and visualization. Seowhi Choo contributed toward methodology, software, and visualization. Seok Ju Lee contributed toward investigation and methodology. Kyung Woo Park contributed toward methodology. Kyoung Yeon Kim contributed toward Review & editing. Seok Hyun Youn contributed toward conceptualization, writing—review & editing, funding acquisition, and supervision.

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Data availability The data underlying this article will be shared on reasonable request to the corresponding author.

Declarations

Conflict of interest There is no conflict of interest to declare.

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References

Baker, M. E., King, R. S., & Kahle (2023). TITAN2: threshold indicator taxa analysis. R package version 2.4.3. https://CRAN.R-project.org/package=TITAN2.

Banse, K., 1995. Zooplankton: pivotal role in the control of ocean production: I. Biomass and production. ICES Journal of Marine Science 52: 265–277. https://doi.org/10.1016/1054-3139(95)80043-3.

Barton, A. D., A. J. Pershing, E. Litchman, N. R. Record, K. F. Edwards, Z. V. Finkel, T. Kiørboe & B. A. Ward, 2013. The biogeography of marine plankton traits. Ecology Letters 16: 522–534. https://doi.org/10.1111/ele.12063.

Batten, S. D. & A. W. Walne, 2011. Variability in northwards extension of warm water copepods in the NE Pacific. Journal of Plankton Research 33: 1643–1653. https://doi.org/10.1093/plankt/fbr065.

Beaufort, L. & M. Grelaud, 2017. A 2700-year record of ENSO and PDO variability from the Californian margin based on coccolithophore assemblages and calcification. Progress in Earth and Planetary Science 4: 5. https://doi.org/10.1186/s40645-017-0123-z.

Beaugrand, G., K. M. Brander, J. Alistair Lindley, S. Souissi & P. C. Reid, 2003. Plankton effect on cod recruitment in the North Sea. Nature 426: 661–664. https://doi.org/10.1038/nature02164.

Beaugrand, G., M. Edwards & L. Legendre, 2010. Marine biodiversity, ecosystem functioning, and carbon cycles. Proceedings of the National Academy of Sciences 107: 10120–10124. https://doi.org/10.1073/pnas.0913855107.

Benedetti, F., S. Gasparini & S. D. Ayata, 2016. Identifying copepod functional groups from species functional traits. Journal of Plankton Research 38: 159–166. https://doi.org/10.1093/plankt/fbv096.

Benedetti, F., M. Vogt, D. Righetti, F. Guilhaumon & S. D. Ayata, 2018. Do functional groups of planktonic copepods differ in their ecological niches? Journal of Biogeography 45: 604–616. https://doi.org/10.1111/jbi.13166.

Bonnet, D. & C. Frid, 2004. Seven copepod species considered as indicators of water-mass influence and changes: results from a Northumberland coastal station. ICES Journal of Marine Science 61: 485–491. https://doi.org/10.1016/j.icesjms.2004.03.005.

Borja, A., 2014. Grand challenges in marine ecosystems ecology. Frontiers in Marine Science 1: 1. https://doi.org/10.3389/fmars.2014.00001.

Boutin, K., S. M. Gaudron, J. Denis & F. B. R. Lasram, 2023. Potential marine benthic colonisers of offshore wind farms in the English Channel: A functional trait-based approach. Marine Environmental Research. https://doi.org/10.1016/j.marenvres.2023.106061.

Bradford-Grieve, J. M., G. A. Boxshall, S. T. Ahyong & S. Ohtsuka, 2010. Cladistic analysis of the calanoid Copepoda. Invertebrate Systematics 24: 291–321. https://doi.org/10.1071/IS10007.

Brun, P., K. Stamieszkin, A. W. Visser, P. Licandro, M. R. Payne & T. Kiørboe, 2019. Climate change has altered zooplankton-fuelled carbon export in the North Atlantic. Nature Ecology and Evolution 3: 416–423. https://doi.org/10.1038/s41559-018-0780-3.

Chavez, F. P., J. Ryan, S. E. Lluch-Cota & M. Niquen, 2003. From anchovies to sardines and back: multidecadal change in the Pacific Ocean. Science 299: 217–221.

Chen, C. T. A. & S. L. Wang, 1999. Carbon, alkalinity and nutrient budgets on the East China Sea continental shelf. Journal of Geophysical Research 104: 20675–20686. https://doi.org/10.1029/1999JC900055.

Chen, C. C., G. C. Gong, W. C. Chou, C. C. Chung, C. H. Hsieh, F. K. Shiah & K. P. Chiang, 2017. The influence of episodic flooding on a pelagic ecosystem in the East China Sea. Biogeosciences 14: 2597–2609. https://doi.org/10.5194/bg-14-2597-2017.

Chihara, M., 1997. An illustrated guide to marine plankton in Japan, Tokai University Press, Tokyo:, 260.

Choi, S. Y., M. C. Jang, S. H. Youn, M. H. Seo & H. Y. Soh, 2021a. Egg production and hatching patterns of *Acartia erythraea* (Copepoda, Calanoida), with a note on its two egg types, in a eutrophic bay in Korea. Journal of Plankton Research 43: 428–441. https://doi.org/10.1093/plankt/fbab030.

Choi, S. Y., E. H. Lee, H. Y. Soh & M. C. Jang, 2021b. Effects of temperature and salinity on egg production, hatching, and mortality rates in *Acartia ohtsukai* (Copepoda, Calanoida). Frontiers in Marine Science 8: 704479. https://doi.org/10.3389/fmars.2021.704479.

Choo, S., M. T. Kwak, Y. K. Cho, Y. H. Yoon & H. Y. Soh, 2023. Effects of water masses on the zooplankton community structure in the northern East China Sea during the East Asian Summer Monsoon in 2020. Ecological Indicators 154: 110847. https://doi.org/10.1016/j.ecolind.2023.110847.

Clarke, A. & K. J. Gaston, 2006. Climate, energy and diversity. Proceedings of the Royal Society B: Biological Sciences 273(1599): 2257–2266. https://doi.org/10.1098/rspb.2006.3545.

Cui, T., X. Chen, X. Zou, Q. Zhang, S. Li & H. Zeng, 2022. State of the climate in the three gorges region of the Yangtze River basin in 2020. Atmospheric and Oceanic Science Letters 15: 100112. https://doi.org/10.1016/j.aosl.2021.100112.

Daufresne, M., K. Lengfellner & U. Sommer, 2009. Global warming benefits the small in aquatic ecosystems. Proceedings of the National Academy of Sciences 106(31): 12788–12793. https://doi.org/10.1073/pnas.0902080106.

Ducklow, H. W., D. K. Steinberg & K. O. Buesseler, 2001. Upper ocean carbon export and the biological pump. Oceanography 14: 50–58.

Dufrêne, M. & P. Legendre, 1997. Species assemblages and indicator species: the need for a flexible asymmetrical approach. Ecological Monographs 67: 345–366.

Edwards, M. & A. J. Richardson, 2004. Impact of climate change on marine pelagic phenology and trophic mismatch. Nature 430: 881–884. https://doi.org/10.1038/nature02808.

Falkenhaug, T., C. Broms, E. Bagøien & N. Nikolioudakis, 2022. Temporal variability of co-occurring *Calanus finmarchicus* and *C. helgolandicus* in Skagerrak. Frontiers in Marine Science 9: 779335. https://doi.org/10.3389/fmars.2022.779335.

Fu, H., J. Zhong, G. Yuan, L. Ni, P. Xie & T. Cao, 2014. Functional traits composition predict macrophytes community productivity along a water depth gradient in a freshwater lake. Ecology and Evolution 4: 1516–1523. https://doi.org/10.1002/ece3.1022.

Gagic, V., I. Bartomeus, T. Jonsson, A. Taylor, C. Winqvist, C. Fischer, E. M. Slade, I. Steffan-Dewenter, M. Emmerson, S. G. Potts, T. Tscharntke & R. Bommarco, 2015. Functional identity and diversity of animals predict ecosystem functioning better than species-based indices. Proceedings of the Royal Society b: Biological Sciences 282: 20142620. https://doi.org/10.1098/rspb.2014.2620.

Giering, S. L., R. Sanders, R. S. Lampitt, T. R. Anderson, C. Tamburini, M. Boutrif, M. V. Zubkov, C. M. Marsay, S. A. Henson, K. Saw, K. Cook & D. J. Mayor, 2014. Reconciliation of the carbon budget in the ocean's twilight zone. Nature 507: 480–483. https://doi.org/10.1038/nature13123.

Gitay, H. & I. R. Noble, 1997. What are functional types and how should we seek them? In Smith, M. M., H. H. Shugart & F. I. Woodward (eds), Plant Functional Types University Press, Cambridge: 3–19.

Harfmann, J., T. Kurobe, B. Bergamaschi, S. Teh & P. Hernes, 2019. Plant detritus is selectively consumed by estuarine copepods and can augment their survival. Scientific Reports 9: 9076. https://doi.org/10.1038/s41598-019-45503-6.

Harvey, B. P., K. E. Marshall, C. D. Harley & B. D. Russell, 2022. Predicting responses to marine heatwaves using functional traits. Trends in Ecology and Evolution 37: 20–29. https://doi.org/10.1016/j.tree.2021.09.003.

Hooff, R. C. & W. T. Peterson, 2006. Copepod biodiversity as an indicator of changes in ocean and climate conditions of the northern California current ecosystem. Limnology and Oceanography 51: 2607–2620. https://doi.org/10.4319/lo.2006.51.6.2607.

Ichikawa, H. & R. C. Beardsley, 2002. The current system in the Yellow and East China Seas. Journal of Oceanography 58: 77–92. https://doi.org/10.1023/A:1015876701363.

Irnawati, I., F. D. O. Riswanto, S. Riyanto, S. Martono & A. Rohman, 2020. The use of software packages of R factoextra and FactoMineR and their application in principal component analysis for authentication of oils. Indonesian Journal of Chemometrics and Pharmaceutical Analysis. https://doi.org/10.22146/ijcpa.482.

Isobe, A., E. Fujiwara, P. H. Chang, K. Sugimatsu, M. Shimizu, T. Matsuno & A. Manda, 2004. Intrusion of less saline shelf water into the Kuroshio subsurface layer in the East China Sea. Journal of Oceanography 60: 853–863. https://doi.org/10.1007/s10872-004-5778-6.

Jansson, A., R. Klais-Peets, E. Grinienė, G. Rubene, A. Semenova, A. Lewandowska & J. Engström-Öst, 2020. Functional shifts in estuarine zooplankton in response to climate variability. Ecology and Evolution 10: 11591–11606. https://doi.org/10.1002/ece3.6793.

Jeong, Y. S., S. Choo & H. Y. Soh, 2022. Influence of rainfall events on zooplankton community characteristics and feeding habits in estuarine–coastal environments. Frontiers in Marine Science 9: 950695. https://doi.org/10.3389/fmars.2022.950695.

Jónasdóttir, S. H., A. W. Visser, K. Richardson & M. R. Heath, 2015. Seasonal copepod lipid pump promotes carbon sequestration in the deep North Atlantic. Proceedings of the National Academy of Sciences 112: 12122–12126. https://doi.org/10.1073/pnas.1512110112.

Jung, S. W., J. G. Park, D. H. Jeong & D. Lim, 2012. Seasonal changes in water masses and phytoplankton communities in the western part of south coastal waters, Korea. Korean Journal of Environmental Biology 30: 328–338. https://doi.org/10.11626/KJEB.2012.30.4.328.

Kim, G. & H. K. Kang, 2020. Mesozooplankton community structure in the Yellow Sea in spring. Ocean and Polar Research 42: 271–281. https://doi.org/10.4217/OPR.2020.42.4.271.

Kim, Y. S., C. J. Jang, J. Y. Jeong & J. S. Shim, 2018. Daily to seasonal variability of the mixed layer depth in the central Yellow Sea: Effects of atmospheric forcing. Journal of Coastal Research 85: 576–580. https://doi.org/10.2112/SI85-116.1.

Kim, G., W. Park, D. H. Choi & H. K. Kang, 2024. Seasonal variation in the mesozooplankton community structure and indicator species of the Yellow Sea. Regional Studies in Marine Science 73: 103454. https://doi.org/10.1016/j.rsma.2024.103454.

Kiørboe, T., 2011. How zooplankton feed: Mechanisms, traits and trade-offs. Biological Reviews 86: 311–339. https://doi.org/10.1111/j.1469-185X.2010.00148.x.

Kiørboe, T., H. Jiang & S. P. Colin, 2010. Danger of zooplankton feeding: The fluid signal generated by ambush-feeding copepods. Proceedings of the Royal Society b: Biological Sciences 277: 3229–3237. https://doi.org/10.1098/rspb.2010.0629.

Kwak, J. H., J. Hwang, E. J. Choy, H. J. Park, D. J. Kang, T. Lee, K. I. Chang, K. R. Kim & C. K. Kang, 2013. High primary productivity and f-ratio in summer in the Ulleung basin of the East/Japan Sea. Deep Sea Research Part i: Oceanographic Research Papers 79: 74–85. https://doi.org/10.1016/j.dsr.2013.05.011.

Kwon, K. Y., J. H. Shim & J. M. Shim, 2019. Temporal variations of sea water environment and nutrients in the east coast of Korea in 2013–2017: Sokcho, Jukbyeon and Gampo Coastal Areas. Journal of the Korean Society of Marine Environment and Safety 25: 457–467. https://doi.org/10.7837/kosomes.2019.25.4.457.

Lee, D. K., & J. C. Lee, 2017. Distribution and circulation of autumn low-salinity water in the East Sea. Korean Journal of Fisheries and Aquatic Sciences 50: 207–218.

Lee, J. Y., D. J. Kang, I. N. Kim, T. Rho, T. Lee, C. K. Kang & K. R. Kim, 2009. Spatial and temporal variability in the pelagic ecosystem of the East Sea (Sea of Japan): A review. Journal of Marine System 78: 288–300. https://doi.org/10.1016/j.jmarsys.2009.02.013.

Lee, S. H., H. Joo, J. H. Lee, J. H. Lee, J. J. Kang, H. W. Lee, D. Lee & C. K. Kang, 2017. Seasonal carbon uptake rates of phytoplankton in the northern East/Japan Sea. Deep Sea Research Part II: Topical Studies in Oceanography 143: 45–53. https://doi.org/10.1016/j.dsr2.2017.04.009.

Lee, J. Y., W. Park, S. Y. Moon & H. K. Cha, 2019. Distribution and indicator species of zooplankton in the East Sea, Yellow Sea and Northern East China Sea in winter. Journal of Environmental Biology 40: 871–883. https://doi.org/10.22438/jeb/40/5(SI)/SI-07.

Li, J., G. Li, J. Xu, P. Dong, L. Qiao, S. Liu, P. Sun & Z. Fan, 2016. Seasonal evolution of the Yellow Sea Cold Water Mass and its interactions with ambient hydrodynamic system. Journal of Geophysical Research: Oceans 121: 6779–6792. https://doi.org/10.1002/2016JC012186.

Li, Y., R. Ge, H. Chen, Y. Zhuang, G. Liu & Z. Zheng, 2022. Functional diversity and groups of crustacean zooplankton in the southern Yellow Sea. Ecological Indicators 136: 108699. https://doi.org/10.1016/j.ecolind.2022.108699.

Lie, H. J. & C. H. Cho, 2016. Seasonal circulation patterns of the Yellow and East China Seas derived from satellite-tracked drifter trajectories and hydrographic observations. Progress in Oceanography 146: 121–141. https://doi.org/10.1016/j.pocean.2016.06.004.

Litchman, E., M. D. Ohman & T. Kiørboe, 2013. Trait-based approaches to zooplankton communities. Journal of Plankton Research 35: 473–484. https://doi.org/10.1093/plankt/fbt019.

Liu, Z., J. Gan, J. Hu, H. Wu, Z. Cai & Y. Deng, 2021. Progress on circulation dynamics in the East China Sea and southern Yellow Sea: Origination, pathways, and destinations of shelf currents. Progress in Oceanography 193: 102553. https://doi.org/10.1016/j.pocean.2021.102553.

Mackas, D. L., S. Batten & M. Trudel, 2007. Effects on zooplankton of a warmer ocean: recent evidence from the Northeast Pacific. Progress in Oceanography 75: 223–252. https://doi.org/10.1016/j.pocean.2007.08.010.

Mackas, D. L., W. Greve, M. Edwards, S. Chiba, K. Tadokoro, D. Eloire, M. G. Mazzocchi, S. Batten, A. J. Richardson, C. Johnson, E. Head, A. Conversi & T. Peluso, 2012. Changing zooplankton seasonality in a changing ocean: Comparing time series of zooplankton phenology. Progress in Oceanography 97–100: 31–62. https://doi.org/10.1016/j.pocean.2011.11.005.

Mackas, D., M. Galbraith, D. Faust, D. Masson, K. Young, W. Shaw, S. Romaine, M. Trudel, J. Dower, R. Campbell, A. Sastri, E. A. Bornhold Pechter, E. Pakhomov & R. El-Sabaawi, 2013. Zooplankton time series from the Strait of Georgia: Results from year-round sampling at deep water locations, 1990–2010. Progress in Oceanography 115: 129–159. https://doi.org/10.1016/j.pocean.2013.05.019.

Malzahn, A. M., F. Hantzsche, K. L. Schoo, M. Boersma & N. Aberle, 2010. Differential effects of nutrient-limited primary production on primary, secondary or tertiary consumers. Oecologia 162: 35–48. https://doi.org/10.1007/s00442-009-1458-y.

Martinez, A. P., 2017. Pairwiseadonis: Pairwise multilevel comparison using Adonis. R package version 1.

McGinty, N., A. D. Barton, N. R. Record, Z. V. Finkel & A. J. Irwin, 2018. Traits structure copepod niches in the North Atlantic and Southern Ocean. Marine Ecology Progress Series 601: 109–126. https://doi.org/10.3354/meps12660.

McGinty, N., A. D. Barton, N. R. Record, Z. V. Finkel, D. G. Johns, C. A. Stock & A. J. Irwin, 2021. Anthropogenic climate change impacts on copepod trait biogeography. Global Change Biology 27: 1431–1442. https://doi.org/10.1111/gcb.15499.

Menéndez, M. C., A. L. Delgado, A. A. Berasategui, M. C. Piccolo & M. S. Hoffmeyer, 2016. Seasonal and tidal dynamics of water temperature, salinity, chlorophyll-*a*, suspended particulate matter, particulate organic matter, and zooplankton abundance in a shallow, mixed estuary (Bahía Blanca, Argentina). Journal of Coastal Research 32: 1051–1061. https://doi.org/10.2112/JCOASTRES-D-14-00236.1.

Miller, C. B., J. A. Crain & C. A. Morgan, 2000. Oil storage variability in *Calanus finmarchicus*. ICES Journal of Marine Science 57(6): 1786–1799. https://doi.org/10.1006/jmsc.2000.0975.

Mitra, A., C. Castellani, W. C. Gentleman, S. H. Jónasdóttir, K. J. Flynn, A. Bode, C. Halsband, P. Kuhn, P. Licandro, M. D. Agersted, A. Calbet & M. S. John, 2014. Bridging the gap between marine biogeochemical and fisheries sciences: Configuring the zooplankton link. Progress in Oceanography 129: 176–199. https://doi.org/10.1016/j.pocean.2014.04.025.

Moretti, M., A. T. Dias, F. De Bello, F. Altermatt, S. L. Chown, F. M. Azcárate, J. R. Bell, B. Fournier, M. Hedde, J. Hortal, S. Ibanez & M. P. Berg, 2017. Handbook of protocols for standardized measurement of terrestrial invertebrate functional traits. Functional Ecology 31: 558–567. https://doi.org/10.1111/1365-2435.12776.

Morgan, C. A., A. De Robertis & R. W. Zabel, 2005. Columbia River plume fronts. I. Hydrography, zooplankton distribution, and community composition. Marine Ecology Progress Series 299: 19–31. https://doi.org/10.3354/meps299019.

Mouillot, D., N. A. Graham, S. Villéger, N. W. Mason & D. R. Bellwood, 2013. A functional approach reveals community responses to disturbances. Trends in Ecology and Evolution 28: 167–177. https://doi.org/10.1016/j.tree.2012.10.004.

Nandini, S. & S. S. S. Sarma, 2003. Population growth of some genera of cladocerans (Cladocera) in relation to algal food (*Chlorella vulgaris*) levels. Hydrobiologia 491: 211–219. https://doi.org/10.1023/A:1024410314313.

Oksanen, J., G. L. Simpson, F. G. Blanchet, R. Kindt, P. Legendre, P. R. Minchin, … & J. Weedon, 2022. vegan: Community Ecology Package. R package version 2.6–4.

Park, K., J. E. Park, B. J. Choi, D. S. Byun & E. I. Lee, 2013. An oceanic current map of the East Sea for science textbooks based on scientific knowledge acquired from oceanic measurements. Journal of Korean Society of Oceanography 18: 234–265. https://doi.org/10.7850/jkso.2013.18.4.234.

Park, K. W., H. J. Oh, S. Y. Moon, M. H. Yoo & S. H. Youn, 2022. Effects of miniaturization of the summer phytoplankton community on the marine ecosystem in the northern East China Sea. Journal of Marine Science and Engineering 10: 315. https://doi.org/10.3390/jmse10030315.

Petchey, O. L. & K. J. Gaston, 2002. Functional diversity (FD), species richness and community composition. Ecology Letters 5: 402–411. https://doi.org/10.1046/j.1461-0248.2002.00339.x.

Pierson, J., E. Camatti, R. Hood, T. Kogovšek, D. Lučić, V. Tirelli & A. Malej, 2020. Mesozooplankton and gelatinous zooplankton in the face of environmental stressors. Coastal Ecosystems in Transition: A Comparative Analysis of the Northern Adriatic and Chesapeake Bay, pp. 105–127. https://doi.org/10.1002/9781119543626.ch6.

Pitois, S. G. & C. J. Fox, 2006. Long-term changes in zooplankton biomass concentration and mean size over the Northwest European shelf inferred from Continuous Plankton Recorder data. ICES Journal of Marine Science 63: 785–798. https://doi.org/10.1016/j.icesjms.2006.03.009.

Pomerleau, C., A. R. Sastri & B. E. Beisner, 2015. Evaluation of functional trait diversity for marine zooplankton communities in the Northeast subarctic Pacific Ocean. Journal of Plankton Research 37: 712–726. https://doi.org/10.1093/plankt/fbv045.

Protopapa, M., R. Koppelmann, S. Zervoudaki, C. Wunsch, J. Peters, C. Parinos, F. Paraschos, A. Gogou & C. Möllmann, 2019. Trophic positioning of prominent copepods in the epi- and mesopelagic zone of the ultra-oligotrophic eastern Mediterranean Sea. Deep Sea Research Part II: Topical Studies in Oceanography 164: 144–155. https://doi.org/10.1016/j.dsr2.2019.04.011.

R Core Team 2024. R: a language and environment for statistical computing. Found. Stat. Comput. Vienna, Austria. URL https://www.R-project.org/

Record, N. R., A. J. Pershing & F. Maps, 2013. Emergent copepod communities in an adaptive trait-structured model. Ecological Modelling 260: 11–24. https://doi.org/10.1016/j.ecolmodel.2013.03.018.

Ren, S., J. Xie & J. Zhu, 2014. The roles of different mechanisms related to the tide-induced fronts in the Yellow Sea in summer. Advances in Atmospheric Sciences 31: 1079–1089. https://doi.org/10.1007/s00376-014-3236-y.

Richardson, A. J., 2008. In hot water: zooplankton and climate change. ICES Journal of Marine Science 65: 279–295. https://doi.org/10.1093/icesjms/fsn028.

Roman, M. R., S. B. Brandt, E. D. Houde & J. J. Pierson, 2019. Interactive effects of hypoxia and temperature on coastal pelagic zooplankton and fish. Frontiers in Marine Science 6: 139. https://doi.org/10.3389/fmars.2019.00139.

Schmidt, K., A. J. Birchill, A. Atkinson, R. J. Brewin, J. R. Clark, A. E. Hickman, D. G. Johns, M. C. Lohan, A. Milne, S. Pardo, S. Pardo, L. S. Polimene & J. Ussher, 2020. Increasing picocyanobacteria success in shelf waters contributes to long-term food web degradation. Global Change Biology 26: 5574–5587. https://doi.org/10.1111/gcb.15161.

Shi, Y., M. Niu, T. Zuo, J. Wang, Q. Luan, J. Sun, W. Yuan, X. Shan & E. A. Pakhomov, 2020. Inter-annual and seasonal variations in zooplankton community structure in the Yellow Sea with possible influence of climatic variability. Progress in Oceanography 185: 102349. https://doi.org/10.1016/j.pocean.2020.102349.

Shin, S. S., S. Y. Choi, M. H. Seo, S. J. Lee, H. Y. Soh & S. H. Youn, 2022. Spatiotemporal distribution characteristics of copepods in the water masses of the northeastern East China Sea. Journal of Marine Science and Engineering 10: 754. https://doi.org/10.3390/jmse1006075.

Sodré, E. D. O. & R. L. Bozelli, 2019. How planktonic microcrustaceans respond to environment and affect ecosystem: a functional trait perspective. International Aquatic Research 11: 207–223. https://doi.org/10.1007/s40071-019-0233-x.

Soh, H. Y. & S. Y. Moon, 2014. Invertebrate fauna of Korea: marine planktonic copepods. National Institute of Biological Resources, Incheon, 138 p

Soh, H. Y., S. Y. Moon & J. H. Wi, 2013. Invertebrate fauna of Korea: marine planktonic copepods. National Institute of Biological Resources, Incheon, 157 p

Steinberg, D. K. & M. R. Landry, 2017. Zooplankton and the ocean carbon cycle. Annual Review of Marine Science 9: 413–444. https://doi.org/10.1146/annurev-marine-010814-015924.

Strand, E., E. Bagøien, M. Edwards, C. Broms & T. Klevjer, 2020. Spatial distributions and seasonality of four *Calanus* species in the Northeast Atlantic. Progress in Oceanography 185: 102344. https://doi.org/10.1016/j.pocean.2020.102344.

Suikkanen, S., S. Pulina, J. Engström-Öst, M. Lehtiniemi, S. Lehtinen & A. Brutemark, 2013. Climate change and eutrophication induced shifts in northern summer plankton communities. PLoS ONE 8(6): e66475. https://doi.org/10.1371/journal.pone.0066475.

Sunday, J. M., A. E. Bates & N. K. Dulvy, 2011. Global analysis of thermal tolerance and latitude in ectotherms. Proceedings of the Royal Society B: Biological Sciences 278(1713): 1823–1830. https://doi.org/10.1098/rspb.2010.1295.

Tang, Q., J. Yang & D. Sun, 2022. Functional diversity of copepod assemblages along a basin-scale latitudinal gradient in the North Pacific Ocean. Ecological Indicators

141: 109112. https://doi.org/10.1016/j.ecolind.2022.109112.

Taylor, A. H., J. I. Allen & P. A. Clark, 2002. Extraction of a weak climatic signal by an ecosystem. Nature 416: 629–632. https://doi.org/10.1038/416629a.

Tilman, D., 2001. Functional diversity. In Levin, S. A. (ed), Encyclopedia of Biodiversity Academic Press, San Diego: 109–120.

van der Meeren, T., R. E. Olsen, K. Hamre & H. J. Fyhn, 2008. Biochemical composition of copepods for evaluation of feed quality in production of juvenile marine fish. Aquaculture 274: 375–397. https://doi.org/10.1016/j.aquaculture.2007.11.041.

Vilgrain, L., F. Maps, M. Picheral, M. Babin, C. Aubry, J. O. Irisson & S. D. Ayata, 2021. Trait-based approach using in situ copepod images reveals contrasting ecological patterns across an Arctic ice melt zone. Limnology and Oceanography 66: 1155–1167. https://doi.org/10.1002/lno.11672.

Violle, C., M. L. Navas, D. Vile, E. Kazakou, C. Fortunel, I. Hummel & E. Garnier, 2007. Let the concept of trait be functional! Oikos 116: 882–892. https://doi.org/10.1111/j.0030-1299.2007.15559.x.

Vogt, R. J., P. R. Peres-Neto & B. E. Beisner, 2013. Using functional traits to investigate the determinants of crustacean zooplankton community structure. Oikos 122: 1700–1709. https://doi.org/10.1111/j.1600-0706.2013.00039.x.

Wang, R., T. Zuo & K. E. Wang, 2003. The Yellow Sea cold bottom water—an oversummering site for *Calanus sinicus* (Copepoda, Crustacea). Journal of Plankton Research 25: 169–183. https://doi.org/10.1093/plankt/25.2.169.

Wickham, H., 2007. Reshaping Data with the reshape Package. Journal of Statistical Software 21: 1–20.

Wickham, H., 2016. Ggplot2: Elegant Graphics for Data Analysis, Spring-Verlag, New York:

Wickham, H., M. Averick, J. Bryan, W. Chang, L. D. McGowan, R. François, G. Grolemund, A. Hayes, L. Henry, J. Hester, M. Kuhn & H. Yutani, 2019. Welcome to the tidyverse. Journal of Open Source Software 4: 1686. https://doi.org/10.21105/joss.01686.

Xiong, T. Q., Q. S. Wei, W. D. Zhai, C. L. Li, S. Y. Wang, Y. X. Zhang, S. J. Liu & S. Q. Yu, 2020. Comparing subsurface seasonal deoxygenation and acidification in the Yellow Sea and northern East China Sea along the north-to-south latitude gradient. Frontiers in Marine Science 7: 686. https://doi.org/10.3389/fmars.2020.00686.

Xu, L. L., D. X. Wu, X. P. Lin & C. Ma, 2009. The study of the Yellow Sea Warm Current and its seasonal variability. Journal of Hydrodynamics 21: 159–165. https://doi.org/10.1016/S1001-6058(08)60133-X.

Yang, H. W., Y. K. Cho, G. H. Seo, S. H. You & J. W. Seo, 2014. Interannual variation of the southern limit in the Yellow Sea Bottom Cold Water and its causes. Journal of Marine Systems 139: 119–127. https://doi.org/10.1016/j.jmarsys.2014.05.007.

Yin, J., Z. Zhao, G. Zhang, S. Wang & A. Wan, 2013. Tempo-spatial variation of nutrient and chlorophyll-*a* concentrations from summer to winter in the Zhangzi Island Area (Northern Yellow Sea). Journal of Ocean University of China 12: 373–384. https://doi.org/10.1007/s11802-013-2101-4.

Yoon, S. T., K. I. Chang, H. Na & S. Minobe, 2016. An east-west contrast of upper ocean heat content variation south of the subpolar front in the East/Japan Sea. Journal of Geophysical Research 121: 6418–6443. https://doi.org/10.1002/2016JC011891.

York, A., 2018. Marine biogeochemical cycles in a changing world. Nature Reviews Microbiology 16: 259–259. https://doi.org/10.1038/nrmicro.2018.40.

Zhang, S. W., Q. Y. Wang, Y. Lü, H. Cui & Y. L. Yuan, 2008. Observation of the seasonal evolution of the Yellow Sea Cold Water Mass in 1996–1998. Continental Shelf Research 28: 442–457. https://doi.org/10.1016/j.csr.2007.10.002.

Zhong, F., R. Xiang, Y. Yang & M. Zhao, 2018. Evolution of the Southern Yellow Sea Cold Water Mass during the last 7 kyr from benthic foraminiferal evidence. Science China Earth Sciences 61: 1406–1418. https://doi.org/10.1007/s11430-017-9193-6.

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